7 Jacobi polynomials

7.1 Taylor series and Jacobi series

Jacobi polynomials are generalizations of the Gegenbauer polynomials discussed in Section 6.7. Their most important property, from the standpoint of real variables, is pairwise orthogonality on a finite interval of the real line; as a result of this orthogonality, it is feasible to expand in series of Jacobi polynomials any function which is square integrable on the interval with respect to a suitable weight function. For example a Fourier cosine series, being equivalent to a series of Chebyshev polynomials (see Section 6.7), is a special Jacobi series.

From the standpoint of complex variables the situation is somewhat different and technically simpler. A function analytic on a finite line segment in the plane can be expanded in Jacobi series without using the orthogonality property, and indeed there are Jacobi polynomials which are not orthogonal but can serve for expansion. Deferring orthogonality to Section 7.8, we shall take a novel point of view and regard Jacobi series as generalizations of Taylor series. A property called biorthogonality will be the cornerstone of our discussion.

We begin by reviewing some familiar facts about Taylor series in a notation chosen with an eye to later generalization. Let $r \in \mathbb{C}$ and define

$$p_n(x) = (x - r)^n, \qquad q_n(x) = (x - r)^{-n-1}, \qquad n \in \mathbb{N}. \tag{1}$$

Consider a power series with complex coefficients,

$$\sum_{n=0}^{\infty} a_n p_n(x), \tag{2}$$

and define σ by

$$\frac{1}{\sigma} = \limsup_{n\to\infty} |a_n|^{1/n}. \tag{3}$$

For every $\rho \geq 0$, let

$$\begin{aligned} \Omega_\rho &= \{x : |x - r| < \rho\}, \qquad \Lambda_\rho = \{x : |x - r| = \rho\}, \\ A_\rho &= \{x : |x - r| > \rho\}, \qquad \bar{\Omega}_\rho = \Omega_\rho \cup \Lambda_\rho, \\ &\qquad \bar{A}_\rho = A_\rho \cup \Lambda_\rho. \end{aligned} \tag{4}$$

The root test shows that the series (2) converges absolutely on the open circular disk Ω_σ and diverges on the open annulus A_σ. The test fails on Λ_σ, called the circle of convergence. On the closed disk $\bar{\Omega}_\rho$, $p_n(x)$ is bounded by ρ^n. If $\rho < \sigma$, then $\sum |a_n| \rho^n$ converges by the root test, and hence (2) converges uniformly on $\bar{\Omega}_\rho$ by the Weierstrass M test. It follows that the sum of the series (2) is holomorphic on Ω_ρ for every $\rho < \sigma$ and consequently on Ω_σ.

Conversely, if ρ is fixed and f is holomorphic on Ω_ρ, f can be represented by a series of the form (2). Following Cauchy's proof of 1831, if $x \in \Omega_\rho$, we choose τ so that $|x - r| < \tau < \rho$ and define the contour γ to be Λ_τ described counterclockwise. If $y \in \Lambda_\tau$, we expand the Cauchy kernel, $(y - x)^{-1} = [(y - r) - (x - r)]^{-1}$, in a geometric series,

$$\frac{1}{y - x} = \sum_{n=0}^{\infty} p_n(x) q_n(y). \tag{5}$$

Since $|p_n(x) q_n(y)| = |x - r|^n \tau^{-n-1}$ and $|x - r| < \tau$, (5) converges uniformly in y on γ. Inserting (5) in Cauchy's integral formula and integrating term by term, we find

$$f(x) = \frac{1}{2\pi i} \int_\gamma \frac{f(y)}{y - x}\, dy = \sum_{n=0}^{\infty} a_n p_n(x), \tag{6}$$

$$a_n = \frac{1}{2\pi i} \int_\gamma f(y) q_n(y)\, dy. \tag{7}$$

A second use of Cauchy's integral formula on the right side of (7) yields

$$a_n = \frac{1}{n!} f^{(n)}(r). \tag{8}$$

This proves that the Taylor series of f converges to f on Ω_ρ. It follows that the Taylor series of f about r has a circle of convergence which passes through the singularity of f that is closest to r. (If the circle of convergence were larger, the sum of the series would provide a holomorphic continuation of f to a region on which f is not holomorphic.)

The definitions (1) imply that

$$\frac{1}{2\pi i}\int_\gamma p_m(y)q_n(y)\,dy = \delta_{mn}, \tag{9}$$

where the right side is the Kronecker delta. One says that the p's and q's are biorthogonal on γ (and on any simple closed contour encircling the point r).

Suppose that f can be represented by a second series $\sum b_m p_m(x)$. Let γ be a concentric circle inside its circle of convergence. For every y on γ,

$$\sum a_m p_m(y) = \sum b_m p_m(y),$$

and both series converge uniformly on γ. Multiplying by $q_n(y)$ and integrating term by term around γ, we find from (9) that $a_n = b_n$ for every $n \in \mathbb{N}$. Hence the coefficients in (6) are unique.

A Taylor series is an expansion about a single point r, and we will now proceed to consider expansions about a pair of points. We shall denote the line segment joining two points, r and s, in the complex plane by

$$[r, s] = \{z \in \mathbb{C} : z = ur + (1-u)s, 0 \le u \le 1\}. \tag{10}$$

In (1) we replace $x - r$ by $x - [ur + (1-u)s] = u(x-r) + (1-u)(x-s)$. Integrating over [0, 1] with respect to a Dirichlet measure, we obtain the Dirichlet average $R_n(b_1, b_2; x-r, x-s)$ from p_n; that is, the monomials of a Taylor series are being averaged as the center of the expansion runs over a line segment. Similarly, from q_n we obtain an averaged Cauchy kernel of the kind discussed in Section 5.11. Since we have studied in Theorem 6.2-2 and Section 6.3 the analytic continuation of Dirichlet averages with respect to the parameters, we may choose almost any complex b_1 and b_2. Preservation of the biorthogonality of the p's and q's dictates, as we shall see in Section 7.2, the following choice.

DEFINITION 7.1-1 Let $n \in \mathbb{N}$, $(1+\alpha, 1+\beta) \in U_2$, and $x, r, s \in \mathbb{C}$. We define

$$\begin{aligned} p_n(x) &= R_n(-\alpha - n, -\beta - n; x-r, x-s), && x \in \mathbb{C},\\ q_n(x) &= R_{-n-1}(1+\alpha+n, 1+\beta+n; x-r, x-s), && x \in \mathbb{C} - [r, s]. \end{aligned} \tag{11}$$

Remark Note that $(1+\alpha, 1+\beta) \in U_2$ implies $\alpha + \beta \neq -2, -3, \ldots$ and $(-\alpha-\beta-2n, n) \neq 0$.

We shall call p_n a Jacobi polynomial and q_n a Jacobi function of the second kind. The p's and q's satisfy adjoint differential equations (Exercise 7.1-8), whereas the customary functions of the second kind (Exercise 7.1-4) satisfy the same differential equation as the p's. Like the special case (6.10-16), p_n has exact degree n and is monic. If $\alpha = \beta$, it is a Gegenbauer polynomial (often called an ultraspherical polynomial) because of (6.10-12). If α or β is $\pm\frac{1}{2}$, it is a Gegenbauer polynomial by Exercise 6.9-8. By Corollary 6.3-5, q_n is holomorphic on the plane cut along the line segment $[r, s]$. For large values of x it has a Laurent series (Exercise 7.1-5) with leading term x^{-n-1} and therefore may also be called monic. If $r = s$, (11) reduces to (1) by (5.2-2).

The behavior of p_n and q_n for large n is in first approximation independent of α and β, which occur in (11) only in the combinations $\alpha + n$ and $\beta + n$. This is why the main convergence properties of Jacobi series are independent of α and β. We shall see that a series of the form (2), with p_n now defined by (11), has an ellipse of convergence with foci r and s, the ellipse reducing to a circle if $r = s$. If f is holomorphic on an open elliptic disk Ω with foci r and s, and if γ is a confocal ellipse in Ω, we shall find that (5)–(7) and (9) remain valid. Equation (8) must be replaced by

$$a_n = \frac{1}{n!} F^{(n)}(1 + \alpha + n, 1 + \beta + n; r, s); \tag{12}$$

that is, a_n is now a Dirichlet average of the Taylor coefficient over the interfocal line segment. From (12) we can write at once the Jacobi coefficients of some elementary functions, and the general theory of Dirichlet averages can be applied to the a's. If $r = s$, (12) reduces to (8) and we recover Taylor's series as a special case.

If $r \neq s$, so that the interfocal line segment $[r, s]$ has strictly positive length, and if α and β are suitably restricted, the p's will be shown to have an orthogonality property on $[r, s]$. This is a consequence of the more general biorthogonality (9), which holds without restriction on r, s, α, β provided only that $(1 + \alpha, 1 + \beta) \in U_2$.

7.2 Biorthogonality

The proof that Jacobi functions are biorthogonal motivates the choice of parameters in (7.1-11). The correct reduction when $r = s$ is ensured by assuming only that

$$\begin{aligned} p_m(x) &= R_m(\delta, \delta'; x - r, x - s), \\ q_n(x) &= R_{-n-1}(\varepsilon, \varepsilon'; x - r, x - s), \qquad m, n \in \mathbb{N}, \end{aligned} \tag{1}$$

where $(\delta + \delta', m) \neq 0$ and $(\varepsilon, \varepsilon') \in U_2$. Leaving open the possibility that δ, δ' may depend on m and $\varepsilon, \varepsilon'$ on n, we wish to choose these parameters so that the biorthogonality relation (7.1-9) is satisfied for all m and n. For present purposes γ may be any positively oriented rectifiable Jordan curve which contains the line segment $[r, s]$ in its inner region.

It follows from (6.3-4) that

$$\frac{1}{2\pi i}\int_\gamma y^m q_n(y)\,dy = \delta_{mn}, \qquad m \le n, \tag{2}$$

for the nth derivative of y^m is zero if $m < n$ and $m!$ if $m = n$. Since $p_m(y)$ is a polynomial of degree m with unit coefficient of y^m, no further restrictions on $\delta, \delta', \varepsilon, \varepsilon'$ are needed to guarantee that (7.1-9) holds for $m \le n$.

We turn next to the case $m > n$. By (6.4-2),

$$p_m(y) = \sum_{k=0}^{m} \binom{m}{k} y^{m-k} R_k(\delta, \delta'; -r, -s), \tag{3}$$

and by (6.3-4),

$$\frac{1}{2\pi i}\int_\gamma y^{m-k} q_n(y)\,dy = \binom{m-k}{n} R_{m-k-n}(\varepsilon, \varepsilon'; r, s). \tag{4}$$

Since the binomial coefficient vanishes if $m - k < n$, (3) and (4) imply

$$\frac{1}{2\pi i}\int_\gamma p_m(y) q_n(y)\,dy$$
$$= \frac{m!}{n!}\sum_{k=0}^{m-n} \frac{1}{k!\,(m-n-k)!} R_k(\delta, \delta'; -r, -s) R_{m-n-k}(\varepsilon, \varepsilon'; r, s). \tag{5}$$

By (5.8-4) the sum on the right side is the term which is homogeneous of degree $m - n$ in r and s in the product

$$S(\delta, \delta'; -r, -s) S(\varepsilon, \varepsilon'; r, s) = {}_1F_1(\delta; \delta + \delta'; s - r)\, {}_1F_1(\varepsilon; \varepsilon + \varepsilon'; r - s).$$

In the last step we have used (5.8-6). By Exercise 2.4-9 the term in question is

$$\frac{(\delta, m-n)(r-s)^{m-n}}{(\delta+\delta', m-n)(m-n)!}\, {}_3F_2\left[\begin{matrix} -m+n,\ 1-\delta-\delta'-m+n,\ \varepsilon \\ 1-\delta-m+n,\ \varepsilon+\varepsilon' \end{matrix};1\right]. \tag{6}$$

We need now to choose the parameters so that the terminating ${}_3F_2$ series can be summed and its sum is zero. If $\varepsilon = 1 - \delta - m + n$ one of the numerator parameters cancels one of the denominator parameters and the series reduces to

$${}_2F_1(-m+n, \varepsilon - \delta'; \varepsilon + \varepsilon'; 1) = \frac{(\varepsilon' + \delta', m-n)}{(\varepsilon + \varepsilon', m-n)}.$$

The right side, obtained from Exercise 2.4-2, vanishes for $m > n$ if we choose $\varepsilon' + \delta' = 1 - m + n$. (If we put $\varepsilon' + \delta' = 0$ the symmetry between primed and unprimed parameters would be lost and the points r and s would no longer be treated on an equal footing.) In summary we have chosen

$$-\delta - m = \varepsilon - 1 - n, \qquad -\delta' - m = \varepsilon' - 1 - n. \tag{7}$$

The left side of the first equation depends on m but not on n, and the right side on n but not on m. If equality is to hold for all m and n with $m > n$, each side must equal a constant, say α. Similarly each side of the second equation must equal a constant β, and thus

$$\delta = -\alpha - m, \qquad \varepsilon = 1 + \alpha + n, \qquad \delta' = -\beta - m, \qquad \varepsilon' = 1 + \beta + n. \tag{8}$$

These are just the values in (7.1-11), where p_n is shown instead of p_m.

We have now proved the biorthogonality theorem which is stated in Theorem 7.2-1. It will be used in Section 7.6 to expand the Cauchy kernel in Jacobi series and to prove the uniqueness of the Jacobi series of an analytic function.

THEOREM 7.2-1 Let $m, n \in \mathbb{N}^2$, $(1 + \alpha, 1 + \beta) \in U_2$, and $r, s \in \mathbb{C}$. Let γ be a positively oriented rectifiable Jordan curve which contains $[r, s]$ in its inner region, and define p_m and q_n by (7.1-11). Then

$$\frac{1}{2\pi i}\int_\gamma p_m(y)q_n(y)\,dy = \delta_{mn}. \tag{9}$$

We shall now represent an arbitrary polynomial as a linear combination of Jacobi polynomials and prove that the coefficients have the form asserted in (7.1-12).

THEOREM 7.2-2 Let $(1 + \alpha, 1 + \beta) \in U_2$ and $x, r, s \in \mathbb{C}$. Define the Jacobi polynomial p_m for every $m \in \mathbb{N}$ by (7.1-11). If f is a polynomial of degree n, then

$$f(x) = \sum_{m=0}^{n} \frac{1}{m!} F^{(m)}(1 + \alpha + m, 1 + \beta + m; r, s)p_m(x). \tag{10}$$

Proof If $m \in \mathbb{N}$, the monomial x^m can be written as a finite sum of Jacobi polynomials with constant coefficients,

$$x^m = \sum_{k=0}^{m} A_{mk}\, p_k(x). \tag{11}$$

(Because p_m has exact degree m, the explicit polynomial expressions for the p's in Exercise 7.1-5 can be solved successively for the monomials.) Hence any polynomial is a finite sum of p's, in particular,

$$f(x) = \sum_{k=0}^{n} a_k\, p_k(x). \tag{12}$$

Multiplying by $q_m(x)$, where $0 \le m \le n$, and integrating around a simple closed contour γ which encircles $[r, s]$ in the positive direction, we find from (9) that

$$\frac{1}{2\pi i}\int_\gamma f(y)q_m(y)\,dy = \sum_{k=0}^{n} a_k\,\delta_{km} = a_m. \tag{13}$$

By (7.1-11) and (6.3-4),

$$a_m = \frac{1}{2\pi i}\int_\gamma f(y)R_{-m-1}(1+\alpha+m, 1+\beta+m; y-r, y-s)\,dy$$

$$= \frac{1}{m!}F^{(m)}(1+\alpha+m, 1+\beta+m; r, s). \quad \square \tag{14}$$

In Section 7.6 we shall extend this result so that f can be an analytic function instead of a polynomial. However, the polynomial case will suffice to prove an important addition theorem in the next section.

7.3 The addition theorem for Gegenbauer polynomials

The addition theorem for Legendre polynomials was discovered by Laplace and Legendre in 1782–1785 and extended by Gegenbauer (1874, 1893) to his polynomials C_n^ν. In the course of the proof we shall want the derivatives of

$$C_n^\nu(x) = \frac{2^n(\nu, n)}{n!}R_n\left(\frac{1}{2}-\nu-n, \frac{1}{2}-\nu-n; x+1, x-1\right). \tag{1}$$

By Exercise 5.9-8, if $0 \le m \le n$,

$$\frac{d^m}{dx^m}C_n^\nu(x) = \frac{2^n(\nu, n)}{(n-m)!}R_{n-m}\left(\frac{1}{2}-\nu-n, \frac{1}{2}-\nu-n; x+1, x-1\right)$$

$$= \frac{2^n(\nu, n)}{2^{n-m}(\nu+m, n-m)}C_{n-m}^{\nu+m}(x) = 2^m(\nu, m)C_{n-m}^{\nu+m}(x). \tag{2}$$

We now turn to the theorem itself. If A and B are (possibly complex) constants, the polynomial

$$f(x) = C_n^\nu(A+Bx) \tag{3}$$

can be represented as a sum of Jacobi polynomials by (7.2-10). The coefficients are Dirichlet averages of the derivatives

$$f^{(m)}(x) = 2^m(\nu, m)B^m C_{n-m}^{\nu+m}(A+Bx). \tag{4}$$

Assume for the moment that $1+\alpha,\ 1+\beta \in \mathbb{C}_>$. Since $A + B[ur + (1-u)s] = u(A+Br) + (1-u)(A+Bs)$, we find

$$F^{(m)}(1+\alpha+m, 1+\beta+m; r, s)$$
$$= 2^m(\nu, m)B^m \int_0^1 C_{n-m}^{\nu+m}[u(A+Br) + (1-u)(A+Bs)] \cdot d\mu_{(1+\alpha+m,\, 1+\beta+m)}(u). \tag{5}$$

The integration can be performed by Gegenbauer's product formula (6.11-5) if we choose $1+\alpha = 1+\beta = \nu$, where $\nu \in \mathbb{C}_>$, with the result that

$$F^{(m)}(\nu+m, \nu+m; r, s) = 2^m(\nu, m)B^m C_{n-m}^{\nu+m}(\cos\theta)C_{n-m}^{\nu+m}(\cos\varphi)/C_{n-m}^{\nu+m}(1),$$
$$A + Br = \cos(\theta+\varphi), \qquad A + Bs = \cos(\theta-\varphi). \tag{6}$$

For simplicity we choose $r = -1$ and $s = 1$, so that $A = \cos\theta\cos\varphi$ and $B = \sin\theta\sin\varphi$. By (7.1-11) and (1) these choices of α, β, r, s imply

$$p_m(x) = R_m(1-\nu-m, 1-\nu-m; x+1, x-1) = \frac{m!}{2^m(\nu-\frac{1}{2}, m)} C_m^{\nu-1/2}(x). \tag{7}$$

Substituting (3), (6), and (7) in (7.2-10), we have the addition theorem stated below in (8). We have assumed $\nu \in \mathbb{C}_>$ in the proof, but the left side of (8) is a polynomial in ν by (6.7-21). Hence the right side also must be a polynomial, the factors in the denominator being cancelled by factors in the numerator. An equality between polynomials holds without restriction.

THEOREM 7.3-1 (*Addition theorem*) Let $n \in \mathbb{N}$ and $\nu, \theta, \varphi, x \in \mathbb{C}$. Then

$$C_n^{\nu}(\cos\theta\cos\varphi + x\sin\theta\sin\varphi)$$
$$= \sum_{m=0}^{n} \frac{(\nu, m)\sin^m\theta\sin^m\varphi}{(\nu-\frac{1}{2}, m)C_{n-m}^{\nu+m}(1)} C_{n-m}^{\nu+m}(\cos\theta)C_{n-m}^{\nu+m}(\cos\varphi)C_m^{\nu-1/2}(x). \tag{8}$$

The case $\nu = \frac{1}{2}$ is especially important for applications. By (1), (6.10-18), and Exercise 6.7-1,

$$\sin^m\theta\, C_{n-m}^{1/2+m}(\cos\theta) = \frac{(-1)^m}{2^m(\frac{1}{2}, m)} P_n^{\ m}(\cos\theta), \qquad C_{n-m}^{1/2+m}(1) = \frac{(n+m)!}{(n-m)!\,(2m)!}. \tag{9}$$

Also, by (7) and (6.10-14),

$$\lim_{\nu\to 1/2} \frac{C_m^{\nu-1/2}(\cos\psi)}{(\nu-\frac{1}{2}, m)} = \frac{2^m}{m!} R_m\left(\frac{1}{2}-m, \frac{1}{2}-m; \cos\psi+1, \cos\psi-1\right)$$
$$= \frac{1}{m!}(2-\delta_{m0})\cos m\psi. \tag{10}$$

Substituting in (8) and using (6.10-19), we find (for $n \in \mathbb{N}$ and $\theta, \varphi, \psi \in \mathbb{C}$) the addition theorem for Legendre polynomials, sometimes called the addition theorem for spherical harmonics:

$$
\begin{aligned}
&P_n(\cos\theta\cos\varphi + \sin\theta\sin\varphi\cos\psi) \\
&\quad = P_n(\cos\theta)P_n(\cos\varphi) + 2\sum_{m=1}^{n} \frac{(n-m)!}{(n+m)!} P_n^{\,m}(\cos\theta)P_n^{\,m}(\cos\varphi)\cos m\psi \\
&\quad = \sum_{m=-n}^{n} (-1)^m P_n^{\,m}(\cos\theta)P_n^{-m}(\cos\varphi)e^{im\psi}. \qquad (11)
\end{aligned}
$$

If θ, φ, ψ are real, the argument of the first member is the cosine of the angle between two vectors in $\mathbb{R}^3$, say $\mathbf{r}$ and $\mathbf{r}'$, whose polar angles are respectively θ and φ and whose azimuthal angles differ by ψ. Note that (11) contains (6.11-6). The addition theorem has great importance in applied mathematics because it represents the Legendre polynomial $P_n(\mathbf{r}\cdot\mathbf{r}'/rr')$ in a form which allows the coordinates of $\mathbf{r}$ to be separated from those of $\mathbf{r}'$.

7.4 Asymptotic properties

To discuss Jacobi series we need information about the behavior of p_n and q_n for large n. The main features of this behavior should be independent of α and β, which occur in (7.1-11) only in the combinations $\alpha + n$ and $\beta + n$. We can get some idea of what to expect by examining the simple Chebyshev case ($\alpha = \beta = -\frac{1}{2}$). By (6.10-11) and (6.2-17), p_n is

$$R_n\left(\frac{1}{2} - n, \frac{1}{2} - n; x^2, y^2\right) = \left(\frac{x+y}{2}\right)^{2n} + \left(\frac{x-y}{2}\right)^{2n}, \qquad n - 1 \in \mathbb{N}. \quad (1)$$

The function q_n is defined by (7.1-11) if the arguments of the R function are nonzero and differ in complex phase by less than π. Thus we define

$$W = \{(x, y) \in \mathbb{C}^2 : xy \neq 0, |\mathrm{ph}\, x - \mathrm{ph}\, y| < \pi/2\}. \quad (2)$$

The Chebyshev function of the second kind is given by Lemma 7.4-1.

LEMMA 7.4-1 Let $n \in \mathbb{N}$ and $(x, y) \in W$. Then

$$R_{-n-1}\left(\frac{1}{2} + n, \frac{1}{2} + n; x^2, y^2\right) = \frac{1}{xy}\left(\frac{x+y}{2}\right)^{-2n}. \quad (3)$$

Proof For the moment assume further that $|\mathrm{ph}\, x| < \pi/4$ and $|\mathrm{ph}\, y| < \pi/4$. By (6.10-1) and (5.9-22),

$$
\begin{aligned}
R_{-n-1}\left(\frac{1}{2} + n, \frac{1}{2} + n; x^2, y^2\right) &= R_{-n-1}\left[n, 1; \left(\frac{x+y}{2}\right)^2, xy\right] \\
&= \frac{1}{xy}\left(\frac{x+y}{2}\right)^{-2n}. \qquad (4)
\end{aligned}
$$

Since the last member is holomorphic in (x, y) on W, the lemma will be proved by the permanence of functional relations if the first member is holomorphic on W. Now $(x, y) \in W$ implies that x^2 and y^2 lie in a half-plane not containing the origin. By Theorem 5.9-2 the first member of (4) is holomorphic in its arguments on any such half-plane, and we need only show that it is single-valued on W. If (x, y) describes a closed curve in W such that x encircles the origin in $\mathbb{C}$ and returns to its initial value multiplied by $e^{i2\pi}$, then y must do likewise because x and y always differ in phase by less than $\pi/2$. Both arguments of R_{-n-1} are multiplied by $e^{i4\pi}$, and the value of the function is multiplied by $(e^{i4\pi})^{-n-1} = 1$. □

We now proceed to general values of α and β, considering first the polynomials and subsequently the functions of the second kind. It will be convenient to define

$$V = \{(x, y) \in \mathbb{C}^2 : xy \neq 0, x^2 \neq y^2\}. \tag{5}$$

THEOREM 7.4-2 Let $n \in \mathbb{N}$, $(x, y) \in V$, and $\alpha, \beta \in \mathbb{C}$. As $n \to \infty$,

$$R_n(-\alpha - n, -\beta - n; x^2, y^2) \sim C_1\left(\frac{x + y}{2}\right)^{2n} + C_2\left(\frac{x - y}{2}\right)^{2n} \tag{6}$$

uniformly on compact subsets of V. The quantities C_1 and C_2 are nonzero and independent of n but may depend on α, β, x, y (see Exercise 7.4-3).

Remark An asymptotic approximation $f_n(z) \sim g_n(z)$ as $n \to \infty$ is said to hold uniformly in z on a set S if $f_n(z)/g_n(z) \to 1$ uniformly on S.

Proof By Exercise 6.6-1,

$$\frac{(-\alpha - \beta - 2n, n)}{n!} R_n(-\alpha - n, -\beta - n; x^2, y^2)$$
$$= \frac{1}{2\pi i}\int_\gamma (1 - tx^2)^{\alpha+n}(1 - ty^2)^{\beta+n}t^{-n-1}\, dt, \tag{7}$$

where γ is a small circle about 0. The integrand depends on n through $[\varphi(t)]^n$, where

$$\varphi(t) = (1 - tx^2)(1 - ty^2)/t = t^{-1} + tx^2y^2 - x^2 - y^2. \tag{8}$$

Hence φ has zeros at $1/x^2$ and $1/y^2$, a pole at 0, and saddle points (where $\varphi' = 0$) at $\pm 1/xy$. All these critical points are distinct and finite because $(x, y) \in V$. The value of φ at the saddle point $1/xy$ is $-(x - y)^2$ and at $-1/xy$ is $-(x + y)^2$.

We evaluate the integral by the saddle-point method, in which the contour γ is deformed by Cauchy's integral theorem so that it passes over one or both saddle points. For large n the modulus of the integrand goes through a very sharp maximum as t passes over a saddle point, and those parts of the contour which are not close to a saddle point make a negligible contribution to the integral. The contribution from the neighborhood of a saddle point [see Copson (1965, Eq. 36.7)] is the product of three factors: the value of the integrand at the saddle point, $n^{-1/2}$, and a nonzero factor independent of n. Hence the right side of (7) is asymptotically approximated by

$$n^{-1/2}C_1'(-1)^n(x+y)^{2n} + n^{-1/2}C_2'(-1)^n(x-y)^{2n}, \tag{9}$$

where C_1' and C_2' are nonzero and independent of n but may depend on α, β, x, y. Contributions from both saddle points must be included because (1) is a special case. The approximation holds uniformly on compact subsets of V because the critical points of φ remain bounded away from one another in a bounded region of the plane if (x, y) is restricted to a compact subset. The proof of the theorem is completed by using Theorems 3.7-1 and 3.4-1 to show that, as $n \to \infty$,

$$\frac{(-\alpha-\beta-2n, n)}{n!} = (-1)^n \frac{(1+\alpha+\beta, 2n)}{(1+\alpha+\beta, n)n!} \sim (-1)^n(\pi n)^{-1/2}2^{\alpha+\beta+2n}. \tag{10}$$

Use of (9) and (10) in (7) yields (6). ▯

Turning to functions of the second kind, we recall the set W defined by (2).

THEOREM 7.4-3 Let $n \in \mathbb{N}$, $(x, y) \in W$, and $\alpha, \beta \in \mathbb{C}$. As $n \to \infty$,

$$R_{-n-1}(1+\alpha+n, 1+\beta+n; x^2, y^2) \sim C_3\left(\frac{x+y}{2}\right)^{-2n} \tag{11}$$

uniformly on compact subsets of W. The quantity C_3 is nonzero and independent of n but may depend on α, β, x, y (see Exercise 7.4-3).

Proof Choose n sufficiently large that $1+\alpha+n, 1+\beta+n \in \mathbb{C}_>$. By (5.1-1),

$$\begin{aligned} &\mathrm{B}(1+\alpha+n, 1+\beta+n)R_{-n-1}(1+\alpha+n, 1+\beta+n; x^2, y^2) \\ &\quad = \int_0^1 [ux^2 + (1-u)y^2]^{-n-1}u^{\alpha+n}(1-u)^{\beta+n}\,du. \end{aligned} \tag{12}$$

The integrand depends on n through $[\varphi(u)]^n$, where

$$\frac{1}{\varphi(u)} = \frac{x^2}{1-u} + \frac{y^2}{u}. \tag{13}$$

Hence φ has zeros at 0 and 1, a pole at $y^2/(y^2-x^2)$, and saddle points at $y/(y \pm x)$. Since $(x, y) \in W$ implies $x + y \neq 0$, all these critical points are distinct and finite if we assume further that $x \neq y$. The value of φ at the saddle point $y/(y+x)$ is $(x+y)^{-2}$ and at $y/(y-x)$ is $(x-y)^{-2}$.

We evaluate the integral by the saddle-point method, deforming the path $[0, 1]$ so that it passes over one or both saddle points. The special case (3) shows that it does not pass over the saddle point at $y/(y-x)$, and the integral must be asymptotically approximated by

$$n^{-1/2}C_3'(x+y)^{-2n}, \tag{14}$$

where C_3' is nonzero and independent of n but may depend on α, β, x, y. Hence the assumption that $x \neq y$ may be dropped. The approximation holds uniformly on compact subsets of W because the saddle point $y/(y+x)$ remains in a bounded region of the plane and is bounded away from the other critical points if (x, y) is restricted to a compact subset. The proof of the theorem is completed by using Theorems 3.7-1 and 3.4-1 to show that as $n \to \infty$,

$$\mathrm{B}(1+\alpha+n, 1+\beta+n) \sim \pi^{1/2}n^{-1/2}2^{-1-\alpha-\beta-2n}. \tag{15}$$

Use of (14) and (15) in (12) yields (11). □

COROLLARY 7.4-4 Let $n \in \mathbb{N}$, $(1+\alpha,\ 1+\beta) \in U_2$, and $x, r, s \in \mathbb{C}$. Define p_n and q_n by (7.1-11). If $x \in \mathbb{C} - \{r, s\}$, then as $n \to \infty$,

$$p_n(x) \sim C_1\left[\frac{(x-r)^{1/2}+(x-s)^{1/2}}{2}\right]^{2n} + C_2\left[\frac{(x-r)^{1/2}-(x-s)^{1/2}}{2}\right]^{2n} \tag{16}$$

uniformly on compact subsets of $\mathbb{C} - \{r, s\}$. If $x \in \mathbb{C} - [r, s]$ and $|\mathrm{ph}(x-r)^{1/2} - \mathrm{ph}(x-s)^{1/2}| < \pi/2$, then as $n \to \infty$,

$$q_n(x) \sim C_3\left[\frac{(x-r)^{1/2}+(x-s)^{1/2}}{2}\right]^{-2n} \tag{17}$$

uniformly on compact subsets of $\mathbb{C} - [r, s]$. The quantities C_1, C_2, C_3 are nonzero and independent of n but may depend on α, β, $x-r$, $x-s$.

Proof In Theorem 7.4-2 and Theorem 7.4-3 replace x and y by $(x-r)^{1/2}$ and $(x-s)^{1/2}$. The definition (5) requires $r \neq s$, but if $r = s$, then (16) is trivial and holds uniformly on $\mathbb{C} - \{r\}$. □

7.5 Convergence of series

A Taylor series about the point r has a circle of convergence with center r, and a Jacobi series about the points r and s has an ellipse of convergence with foci r and s, as we shall now show. For instance we know already that the generating series (6.7-19) of the Gegenbauer polynomials has an ellipse of convergence with foci -1 and 1.

The ellipse with foci r and s which passes through the point x (where $r, s, x \in \mathbb{C}$) has a major axis of length $|x-r|+|x-s|$. Let ψ be the angle subtended at x by the interfocal line segment. The square of the interfocal distance is

$$\begin{aligned} |r-s|^2 &= |(x-r)-(x-s)|^2 \\ &= |x-r|^2+|x-s|^2-2|x-r|\cdot|x-s|\cos\psi \\ &= (|x-r|+|x-s|)^2-4|x-r|\cdot|x-s|\cos^2(\psi/2). \end{aligned} \tag{1}$$

Hence the minor axis has length $2|x-r|^{1/2}|x-s|^{1/2}\cos(\psi/2)$. Define the mean radius of the ellipse to be the arithmetic mean of the lengths of the semiaxes. Denoting this mean radius by $\mu(x)$ and half the difference of the semiaxes by $\nu(x)$, we find

$$\begin{aligned} 4\mu(x) &= |x-r|+|x-s|+2|x-r|^{1/2}|x-s|^{1/2}\cos(\psi/2), \\ 4\nu(x) &= |x-r|+|x-s|-2|x-r|^{1/2}|x-s|^{1/2}\cos(\psi/2), \end{aligned} \tag{2}$$

or without mentioning ψ,

$$\begin{aligned} \mu(x) &= \max\left\{\left|\frac{(x-r)^{1/2}+(x-s)^{1/2}}{2}\right|^2, \left|\frac{(x-r)^{1/2}-(x-s)^{1/2}}{2}\right|^2\right\}, \\ \nu(x) &= \min\left\{\left|\frac{(x-r)^{1/2}+(x-s)^{1/2}}{2}\right|^2, \left|\frac{(x-r)^{1/2}-(x-s)^{1/2}}{2}\right|^2\right\}. \end{aligned} \tag{3}$$

Note that $16\mu\nu = |r-s|^2$. For example, if $r=-1$ and $s=1$, then $\mu(\cos\theta) = \frac{1}{2}\exp|\text{Im }\theta|$ and $\nu(x) = \frac{1}{2}\exp(-|\text{Im }\theta|)$. If $r=s$, then $\mu(x)=|x-r|$ and $\nu(x)=0$.

THEOREM 7.5-1 Let $n \in \mathbb{N}$, $(1+\alpha, 1+\beta) \in U_2$, and $r, s \in \mathbb{C}$. Define p_n and q_n by (7.1-11). If $x \in \mathbb{C}-[r, s]$, then

$$\lim_{n\to\infty}|p_n(x)|^{1/n} = \mu(x), \qquad \lim_{n\to\infty}|q_n(x)|^{1/n} = \frac{1}{\mu(x)}. \tag{4}$$

Both limits are uniform on compact subsets of $\mathbb{C}-[r, s]$.

Remark If x is an interior point of $[r, s]$, then $\limsup|p_n(x)|^{1/n} = \mu(x)$, but even this may fail (unless α and β are further restricted) at the points r and s, where (7.4-16) is inapplicable. An inequality is given in Exercise 7.5-1.

Proof Since $x \in \mathbb{C}-[r, s]$ implies $\mu(x) > \nu(x)$, one of the quantities in brackets in (7.4-16) is strictly larger in absolute value than the other. Hence $|p_n| \sim |C_i|\mu^n$ for some $i \in \{1, 2\}$. The condition on complex phases accompanying (7.4-17) implies $|q_n| \sim |C_3|\mu^{-n}$. Since C_1, C_2, C_3 are nonzero and independent of n, (4) follows at once. Both limits are uniform on compact subsets of $\mathbb{C} - [r, s]$ because (7.4-16) and (7.4-17) hold uniformly. □

The mean radius of an ellipse cannot be less than one quarter of the interfocal distance. If ρ is a real constant such that $4\rho > |r - s|$, then $\mu(x) = \rho$ is the equation of the ellipse with foci r and s and mean radius ρ. If $r = s$, the ellipse is a circle with center r and radius ρ, and $\mu(x)$ reduces to $|x - r|$. For every $\rho \geq |r - s|/4$ we define, in analogy with (7.1-4),

$$\begin{gathered} \Omega_\rho = \{x : \mu(x) < \rho\}, \qquad \Lambda_\rho = \{x : \mu(x) = \rho\}, \qquad A_\rho = \{x : \mu(x) > \rho\}, \\ \bar{\Omega}_\rho = \Omega_\rho \cup \Lambda_\rho, \qquad \bar{A}_\rho = A_\rho \cup \Lambda_\rho. \end{gathered} \tag{5}$$

We shall call Ω_ρ and $\bar{\Omega}_\rho$ the open and closed elliptic disks with foci r and s and mean radius ρ. Their complements are the closed and open elliptic annuli $\bar{A}_\rho$ and A_ρ, respectively. The ellipse Λ_ρ is the boundary of each of these sets. If $4\rho = |r - s|$, Λ_ρ degenerates to the interfocal line segment $[r, s]$ and Ω_ρ is the null set.

THEOREM 7.5-2 Let $n \in \mathbb{N}$, $\rho \in \mathbb{R}$, $(1 + \alpha, 1 + \beta) \in U_2$, and $r, s \in \mathbb{C}$. Define p_n and q_n by (7.1-11). If $4\rho > |r - s|$, define

$$\hat{p}_n(\rho) = \max_{x \in \bar{\Omega}_\rho} |p_n(x)|, \qquad \hat{q}_n(\rho) = \max_{x \in \bar{A}_\rho} |q_n(x)|. \tag{6}$$

Then

$$\lim_{n \to \infty} [\hat{p}_n(\rho)]^{1/n} = \rho, \qquad \lim_{n \to \infty} [\hat{q}_n(\rho)]^{1/n} = 1/\rho. \tag{7}$$

Remark Clearly $\hat{p}_n$ increases continuously with ρ while $\hat{q}_n$ decreases continuously. Both functions depend also on α, β, r, s. If $r = s$, then $\hat{p}_n(\rho) = \rho^n$ and $\hat{q}_n(\rho) = \rho^{-n-1}$.

Proof Since the modulus of an analytic function cannot have a maximum at an interior point, $|p_n|$ attains the value $\hat{p}_n(\rho)$ at some point of Λ_ρ. Since Λ_ρ is a compact subset of $\mathbb{C} - [r, s]$ and has the equation $\mu(x) = \rho$, $|p_n|^{1/n} \to \rho$ uniformly on Λ_ρ as $n \to \infty$ by Theorem 7.5-1; that is, given $\varepsilon > 0$ we can find an integer N such that $\rho - \varepsilon \leq |p_n|^{1/n} \leq \rho + \varepsilon$ on Λ_ρ for all $n \geq N$. Since $\hat{p}_n(\rho)$ is the maximum of $|p_n|$ on Λ_ρ, it follows that

$$\rho - \varepsilon \leq [\hat{p}_n(\rho)]^{1/n} \leq \rho + \varepsilon, \qquad n \geq N,$$

which implies the first equation of (7). The second equation is proved the same way, for $|q_n|$ attains the value $\hat{q}_n(\rho)$ at some point of Λ_ρ because q_n is holomorphic on A_ρ and $q_n(x) \to 0$ as $|x| \to \infty$. □

Finally we show that Jacobi series of the form $\sum a_n p_n(x)$ and $\sum b_n q_n(x)$ have convergence properties similar to those of Taylor and Laurent series, to which they respectively reduce if $r = s$. In each case the coefficients determine a real number which, if larger than one quarter of the interfocal distance, is the mean radius of an ellipse of convergence. The ellipse separates regions of convergence and divergence, the convergence is uniform on compact subsets, and hence the sum of the series is holomorphic on the region of convergence.

THEOREM 7.5-3 Let $(1 + \alpha, 1 + \beta) \in U_2$ and $x, r, s \in \mathbb{C}$. Let $a_n, b_n \in \mathbb{C}$ for every $n \in \mathbb{N}$; define

$$\frac{1}{\sigma} = \limsup_{n \to \infty} |a_n|^{1/n}, \qquad \tau = \limsup_{n \to \infty} |b_n|^{1/n}, \tag{8}$$

and assume $4\sigma > |r - s|$ and $4\tau > |r - s|$. Define p_n and q_n by (7.1-11). Then $\sum_{n=0}^{\infty} a_n p_n(x)$ converges absolutely on Ω_σ and uniformly on every compact subset of Ω_σ, and the sum of the series is holomorphic on Ω_σ. The series diverges on A_σ. The series $\sum_{n=0}^{\infty} b_n q_n(x)$ converges absolutely on A_τ and uniformly on every compact subset of A_τ, and the sum of the series is holomorphic on A_τ. The series diverges on $\Omega_\tau - [r, s]$, and its terms are not defined on $[r, s]$.

Proof Any compact subset of Ω_σ is contained in $\overline{\Omega}_\rho$ for some ρ such that $4\sigma > 4\rho > |r - s|$. The series $\sum a_n p_n(x)$ is majorized on $\overline{\Omega}_\rho$ by $\sum |a_n| \hat{p}_n(\rho)$, which converges by the root test and (7) because

$$\limsup |a_n \hat{p}_n(\rho)|^{1/n} = \limsup |a_n|^{1/n} \lim [\hat{p}_n(\rho)]^{1/n} = \rho/\sigma < 1.$$

Hence $\sum a_n p_n(x)$ converges absolutely and uniformly on $\overline{\Omega}_\rho$ by the Weierstrass M test. Since the series of polynomials converges uniformly on compact subsets of Ω_σ, the sum of the series is holomorphic on Ω_σ. If $x \in A_\sigma$ the series diverges by the root test and (4) because

$$\limsup |a_n p_n(x)|^{1/n} = \limsup |a_n|^{1/n} \lim |p_n(x)|^{1/n} = \mu(x)/\sigma > 1.$$

The proof for $\sum b_n q_n(x)$ is similar. □

7.6 Jacobi series of an analytic function

To expand an analytic function in a series of Jacobi polynomials, we expand first the kernel of Cauchy's integral formula.

LEMMA 7.6-1 Let $r, s, x \in \mathbb{C}$, $(1 + \alpha, 1 + \beta) \in U_2$, and $y \in \mathbb{C} - [r, s]$. Define p_n and q_n by (7.1-11) and μ by (7.5-3). If $\mu(x) < \mu(y)$, then

$$\frac{1}{y - x} = \sum_{n=0}^{\infty} p_n(x) q_n(y). \tag{1}$$

If $|r - s| < 4\rho < 4\sigma$, the series converges uniformly in (x, y) on $\overline{\Omega}_\rho \times \overline{A}_\sigma$.

Proof If $\mu(x) < \mu(y)$, we can choose $\rho > |r - s|/4$ and σ so that $\mu(x) \leq \rho < \sigma \leq \mu(y)$, which implies $x \in \overline{\Omega}_\rho$ and $y \in \overline{A}_\sigma$. Therefore it suffices to show, for every ρ and σ such that $|r - s| < 4\rho < 4\sigma$, that (1) holds uniformly in (x, y) on $\overline{\Omega}_\rho \times \overline{A}_\sigma$.

Denote the length of the semimajor axis of Λ_σ by a and the center of Λ_σ by $\lambda = (r + s)/2$. Assume for the present that $|y - \lambda| > a$ and $x \in \overline{\Omega}_\rho$. Since $|x - \lambda| < a$,

$$(y - x)^{-1} = [(y - \lambda) - (x - \lambda)]^{-1} = \sum_{m=0}^{\infty} (x - \lambda)^m (y - \lambda)^{-m-1}. \qquad (2)$$

By (7.2-12) and (7.2-13),

$$(x - \lambda)^m = \sum_{n=0}^{m} c_{mn} p_n(x), \qquad c_{mn} = \frac{1}{2\pi i} \int_\gamma (w - \lambda)^m q_n(w)\, dw, \qquad (3)$$

where we choose γ to be Λ_σ described counterclockwise. Since $c_{mn} = 0$ if $n > m$ by (7.2-2), we may replace the upper limit of summation by ∞. Substituting (3) in (2) yields

$$\frac{1}{y - x} = \frac{1}{2\pi i} \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} p_n(x) \int_\gamma \frac{(w - \lambda)^m}{(y - \lambda)^{m+1}} q_n(w)\, dw. \qquad (4)$$

If L denotes the length of γ, the double series is majorized by

$$L \sum_{m=0}^{\infty} \frac{a^m}{|y - \lambda|^{m+1}} \sum_{n=0}^{\infty} \hat{p}_n(\rho) \hat{q}_n(\sigma).$$

The first series converges because $|y - \lambda| > a$, and the second converges by the root test and (7.5-7). Hence the double series in (4) converges absolutely, and the order of summation may be changed to give

$$\frac{1}{y - x} = \frac{1}{2\pi i} \sum_{n=0}^{\infty} p_n(x) \sum_{m=0}^{\infty} \int_\gamma \frac{(w - \lambda)^m}{(y - \lambda)^{m+1}} q_n(w)\, dw. \qquad (5)$$

Since $|w - \lambda| \leq a < |y - a|$ for every w on γ, the series

$$\sum_{m=0}^{\infty} \frac{(w - \lambda)^m}{(y - \lambda)^{m+1}} = \frac{1}{(y - \lambda) - (w - \lambda)} = \frac{1}{y - w}$$

converges uniformly on γ. Performing the summation under the integral sign yields

$$\frac{1}{y - x} = \frac{1}{2\pi i} \sum_{n=0}^{\infty} p_n(x) \int_\gamma \frac{1}{y - w} q_n(w)\, dw = \sum_{n=0}^{\infty} p_n(x) q_n(y). \qquad (6)$$

In the last step we have used Cauchy's integral formula for a function which is holomorphic outside a closed contour and vanishes at infinity (see Exercise 7.6-1). We have now proved (1) if $x \in \overline{\Omega}_\rho$ and $|y - \lambda| > a$.

The right side of (1) is majorized on $\overline{\Omega}_\rho \times \overline{A}_\sigma$ by $\sum \hat{p}_n(\rho)\hat{q}_n(\sigma)$, which converges by the root test and (7.5-7). By the Weierstrass M test the original series converges uniformly in (x, y) on $\overline{\Omega}_\rho \times \overline{A}_\sigma$ and its sum is holomorphic. By the permanence of functional relations (1) continues to hold without the earlier restriction on y. ▯

The expansion of the Cauchy kernel now permits the expansion in Jacobi series of any analytic function by the same method that was used for Taylor series in Section 7.1. Most of the equations are formally unchanged.

THEOREM 7.6-2 Let $r, s \in \mathbb{C}$ and $(1 + \alpha, 1 + \beta) \in U_2$. Let f be holomorphic on an open elliptic disk Ω with foci r and s. Define the monic Jacobi polynomial p_n by (7.1-11). Then

$$f(x) = \sum_{n=0}^{\infty} a_n p_n(x), \qquad x \in \Omega, \tag{7}$$

where a_n is a Dirichlet average of the Taylor series coefficient:

$$a_n = \frac{1}{n!} F^{(n)}(1 + \alpha + n, 1 + \beta + n; r, s), \qquad n \in \mathbb{N}. \tag{8}$$

The series (7) converges absolutely on Ω and uniformly on every compact subset of Ω. The coefficients are unique. The series has an ellipse of convergence whose inner region is the largest elliptic disk with foci r and s to which f can be analytically continued.

Proof Let τ denote the mean radius of Ω. If $x \in \Omega$, choose a real number σ such that $\mu(x) < \sigma < \tau$. Then (1) converges uniformly in y on $\overline{A}_\sigma$. Let the contour γ be Λ_σ described counterclockwise. Substitute (1) in Cauchy's integral formula and integrate term by term to obtain

$$f(x) = \frac{1}{2\pi i} \int_\gamma \frac{f(y)}{y - x}\, dy = \sum_{n=0}^{\infty} a_n p_n(x), \tag{9}$$

$$a_n = \frac{1}{2\pi i} \int_\gamma f(y) q_n(y)\, dy. \tag{10}$$

By (6.3-4), (10) implies (8). Define ρ by

$$\frac{1}{\rho} = \limsup_{n \to \infty} |a_n|^{1/n}.$$

According to Theorem 7.5-3 the series in (7) has an ellipse of convergence Λ_ρ, converges absolutely and uniformly on compact subsets of Ω_ρ, and diverges on A_ρ. Since (7) converges for every $x \in \Omega$, it follows that $\Omega \subset \Omega_\rho$.

Hence the series converges absolutely and uniformly on compact subsets of Ω. The sum of the series is holomorphic on Ω_ρ and provides the analytic continuation of f to Ω_ρ in case $\rho > \tau$. Suppose f could be continued analytically to a still larger confocal elliptic disk, say Ω_ω where $\omega > \rho$. By the first part of the theorem with Ω replaced by Ω_ω, (7) would then hold on Ω_ω, which is impossible because the series diverges on A_ρ.

Suppose f can be represented by another series of Jacobi polynomials, say $f = \sum b_m p_m$, which converges at one point (at least) off the interfocal line segment. By the root test and Theorem 7.5-3 the series has an ellipse of convergence Λ_δ with $|r - s|/4 < \delta \le \rho$. Hence $\sum a_m p_m = \sum b_m p_m$ on Ω_δ, and both series converge uniformly on compact subsets of Ω_δ. Multiplying by q_n, integrating term by term around a confocal ellipse in Ω_δ, and using the biorthogonality relation (7.2-9), we find $a_n = b_n$ for every $n \in \mathbb{N}$. That is, the coefficients are unique. □

7.7 Applications

Since the exponential function is entire, its Jacobi series converges everywhere in the complex plane. A special case of great importance in applied mathematics is the expansion of a plane wave, $\exp(ikr \cos \theta)$, in Legendre polynomials $P_n(\cos \theta)$. Expansion of the same function in Fourier cosine series (i.e. in Chebyshev polynomials) leads to a generating relation for Bessel functions of integral order. More generally, the Fourier cosine series of $f(A + B \cos \theta)$, where A and B are constants and f is analytic, has various applications. The Fourier coefficient of this function is proportional to the integral

$$\int_0^\pi f(A + B \cos \theta) \cos n\theta \, d\theta, \qquad n \in \mathbb{N}, \tag{1}$$

and we shall see how to express the integral as a Dirichlet average of $f^{(n)}$. The Jacobi series of a ${}_0F_1$ function leads to Gegenbauer's addition theorem for Bessel functions and the expansion of a spherical wave in Legendre polynomials.

EXAMPLE 7.7-1 (*Expansion of a plane wave in Legendre polynomials*) Let λ and x be complex numbers and set $f(x) = e^{\lambda x}$ in Theorem 7.6-2. Since $f^{(n)}(x) = \lambda^n f(x)$ and $F(b, z) = S(b, \lambda z)$, we find

$$e^{\lambda x} = \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} S(1 + \alpha + n, 1 + \beta + n; \lambda r, \lambda s) p_n(x). \tag{2}$$

If k, ρ, and θ are any complex numbers, we set $\lambda = ik\rho$, $x = \cos\theta$, $\alpha = \beta = 0$, $r = -1$, and $s = 1$. Using (6.10-13), (5.8-23), and (5.8-24), we find

$$e^{ik\rho\cos\theta} = \sum_{n=0}^{\infty} i^n(2n+1)j_n(k\rho)P_n(\cos\theta), \qquad k, \rho, \theta \in \mathbb{C}. \tag{3}$$

This expansion in spherical Bessel functions and Legendre polynomials is often used in the quantum-mechanical theory of collisions. In that context ρ and θ are spherical polar coordinates and $\exp(ik\rho\cos\theta) = \exp(ikz)$ represents a plane wave with wave number k traveling in the z direction.

EXAMPLE 7.7-2 (*Fourier cosine series*) Let $\theta = \theta' + i\theta''$ and $w = \cos\theta = u + iv$, where θ', θ'', u, v are real. Then $u = \cos\theta' \cosh\theta''$, $v = -\sin\theta' \sinh\theta''$, and

$$\left(\frac{u}{\cosh\theta''}\right)^2 + \left(\frac{v}{\sinh\theta''}\right)^2 = 1.$$

If θ'' is fixed, this is the equation of an ellipse in the w plane with foci ± 1, semiaxis lengths $\cosh\theta''$ and $\sinh|\theta''|$, and mean radius $\frac{1}{2}\exp(|\theta''|)$. If h is a positive number and A and B are complex numbers with $B \neq 0$, it follows that the set

$$\Omega = \{x \in \mathbb{C} : x = A + B\cos\theta, |\mathrm{Im}\,\theta| < h\} \tag{4}$$

is a nonempty open elliptic disk with foci $A \pm B$ and mean radius $\frac{1}{2}|B|e^h$.

Let f be holomorphic on Ω and set $\alpha = \beta = -\frac{1}{2}$, $r = A - B$, $s = A + B$ in Theorem 7.6-2. By (6.10-14), for all θ with $|\mathrm{Im}\,\theta| < h$,

$$f(A + B\cos\theta) = F\left(\frac{1}{2}, \frac{1}{2}; A + B, A - B\right)$$
$$+ 2\sum_{n=1}^{\infty} \frac{B^n}{n!\,2^n} F^{(n)}\left(\tfrac{1}{2} + n, \tfrac{1}{2} + n; A + B, A - B\right)\cos n\theta. \tag{5}$$

Since the series converges uniformly on compact subsets of Ω, we may multiply by $\cos n\theta$ and integrate term by term over the interval $0 \le \theta \le \pi$. The result is

$$\int_0^{\pi} f(A + B\cos\theta)\cos n\theta\, d\theta = \frac{\pi B^n}{n!\,2^n} F^{(n)}\left(\tfrac{1}{2} + n, \tfrac{1}{2} + n; A + B, A - B\right),$$
$$n \in \mathbb{N}. \tag{6}$$

EXAMPLE 7.7-3 (*Fourier series of a plane wave and generating function for Bessel coefficients*) In (5) set $f(x) = e^x$, $A = 0$, and $B = ikr$ to obtain

$$e^{ikr\cos\theta} = J_0(kr) + 2\sum_{n=1}^{\infty} i^n J_n(kr)\cos n\theta, \qquad k, r, \theta \in \mathbb{C}. \tag{7}$$

Since $J_n = (-1)^n J_{-n}$ by (6.9-25), (7) can be rewritten in the form

$$e^{ikr\cos\theta} = \sum_{n=-\infty}^{\infty} i^n J_n(kr)e^{in\theta}. \tag{8}$$

We set $ikr = x$ and use (6.9-22), or alternatively we set $kr = x$ and replace θ by $\theta - \pi/2$. Thus

$$e^{x\cos\theta} = \sum_{n=-\infty}^{\infty} I_n(x)e^{in\theta}, \qquad e^{ix\sin\theta} = \sum_{n=-\infty}^{\infty} J_n(x)e^{in\theta}, \qquad x, \theta \in \mathbb{C}. \tag{9}$$

If $t = e^{i\theta}$, these equations become generating relations, valid for $x \in \mathbb{C}$ and $t \in \mathbb{C} - \{0\}$,

$$\exp\left[\frac{1}{2}x(t + t^{-1})\right] = \sum_{n=-\infty}^{\infty} t^n I_n(x), \qquad \exp\left[\frac{1}{2}x(t - t^{-1})\right] = \sum_{n=-\infty}^{\infty} t^n J_n(x). \tag{10}$$

EXAMPLE 7.7-4 (*Gegenbauer's addition theorem for Bessel functions*) The function $f(x) = {}_0F_1(c; x)$ has fairly simple Jacobi coefficients if $\alpha = \beta = c - \frac{3}{2}$. We require $2c - 1 \in U$ so that $(1 + \alpha, 1 + \beta) \in U_2$. By (6.9-23),

$$f^{(n)}(x) = \frac{1}{(c, n)}\,{}_0F_1(c + n; x) = \frac{1}{(c, n)}\sum_{m=0}^{\infty} \frac{x^m}{(c + n, m)m!},$$

and hence, by (6.3-1) and Exercise 7.1-9,

$$\begin{aligned} F^{(n)}\left(c - \frac{1}{2} + n, c - \frac{1}{2} + n; r, s\right) &= \frac{1}{(c, n)}\sum_{m=0}^{\infty} \frac{1}{(c + n, m)m!} \\ &\quad \cdot R_m(c - \tfrac{1}{2} + n, c - \tfrac{1}{2} + n; r, s) \\ &= \frac{1}{(c, n)}\,{}_0F_1(c + n; t_+)\,{}_0F_1(c + n; t_-), \end{aligned} \tag{11}$$

$$t_\pm = \left(\frac{\sqrt{r} \pm \sqrt{s}}{2}\right)^2, \qquad r, s \in \mathbb{C}.$$

By Theorem 7.6-2,

$$\begin{aligned} {}_0F_1(c; x) = \sum_{n=0}^{\infty} \frac{1}{(c, n)n!}\,{}_0F_1(c + n; t_+)\,{}_0F_1(c + n; t_-) \\ \cdot R_n(\tfrac{3}{2} - c - n, \tfrac{3}{2} - c - n; x - r, x - s). \end{aligned} \tag{12}$$

Change of notation makes this an addition theorem for Bessel functions. Let A and B be complex numbers and replace c by $1 + v$ and $-4x$, $-4r$, $-4s$ by $A^2 + B^2 - 2ABx$, $(A - B)^2$, $(A + B)^2$. By (6.9-18) and (6.10-12),

$$w^{-v}J_v(w) = 2^v\Gamma(v)(AB)^{-v} \sum_{n=0}^{\infty} (v + n)J_{v+n}(A)J_{v+n}(B)C_n^{\,v}(x),$$

$$w = (A^2 + B^2 - 2ABx)^{1/2}, \qquad A, B, x \in \mathbb{C}, \qquad 2v \in U. \tag{13}$$

For the case $v = 0$ see Exercise 7.7-6. The case $v = \frac{1}{2}$ includes an expansion of a spherical wave in spherical Bessel functions and Legendre polynomials. Let $k \in \mathbb{C}$, let $\mathbf{r}$ and $\mathbf{r}'$ be vectors in $\mathbb{R}^3$ with lengths r and r', and let $\cos\theta = \mathbf{r} \cdot \mathbf{r}'/rr'$. Setting $A = kr$, $B = kr'$, $x = \cos\theta$, $v = \frac{1}{2}$ and using (5.8-24) and (6.9-20), we find

$$\frac{\sin k|\mathbf{r} - \mathbf{r}'|}{k|\mathbf{r} - \mathbf{r}'|} = \sum_{n=0}^{\infty} (2n + 1)j_n(kr)j_n(kr')P_n(\cos\theta). \tag{14}$$

7.8 Rodrigues' formula and orthogonality

The Jacobi polynomial $p_n(x)$ can be generated by differentiating an elementary function n times with respect to x. Similarly $R_n(\beta, \beta'; x, y)$ can be generated by partial differentiation (see Exercises 7.8-4 and 7.8-5). Formulas of this kind are called Rodrigues' formulas after the discoverer of their prototype, a formula for Legendre polynomials (Exercise 7.8-1).

FORMULA 7.8-1 (*Rodrigues' formula*) Let $n \in \mathbb{N}$ and $\alpha, \beta, r, s, x \in \mathbb{C}$. If $D = d/dx$, then

$$(1 + \alpha + \beta + n, n)p_n(x) = (x - r)^{-\beta}(x - s)^{-\alpha}D^n[(x - r)^{\beta+n}(x - s)^{\alpha+n}]. \tag{1}$$

Remark Since $x - s$ occurs once on the right side with exponent $-\alpha$ and once with exponent $\alpha + n$, we may replace $x - s$ by $s - x$ in both places if we also multiply the right side by $(-1)^n$. This is convenient if $x \in [r, s]$ and $\mathrm{ph}(s - x) = \mathrm{ph}(x - r)$.

Proof Since the left side of (1) is a polynomial in α and β, it suffices to prove (1) when $(1 + \alpha + \beta + n,\ n) \neq 0$. By (5.9-21) and Exercise 5.9-18,

$$\begin{aligned} p_n(x) &= R_n(-\alpha - n, -\beta - n; x - r, x - s) \\ &= (x - r)^{-\beta}(x - s)^{-\alpha}R_{\alpha+\beta+n}(-\beta - n, -\alpha - n; x - r, x - s) \\ &= \frac{(x - r)^{-\beta}(x - s)^{-\alpha}}{(1 + \alpha + \beta + n, n)} D^n R_{\alpha+\beta+2n}(-\beta - n, -\alpha - n; x - r, x - s). \end{aligned}$$

By (5.9-22) we obtain (1). ◻

Rodrigues' formula is especially useful for integration by parts. We illustrate this by finding another integral representation of the Jacobi coefficient (7.6-8). As a corollary we shall prove the orthogonality of the Jacobi polynomials on the interfocal line segment.

REPRESENTATION 7.8-2 Let $1 + \alpha$, $1 + \beta \in \mathbb{C}_>$ and $r, s \in \mathbb{C}$, and assume $r \neq s$. For every $n \in \mathbb{N}$ define $p_n(x)$ by (7.1-11) and let

$$I_n = \frac{n!\mathrm{B}(1 + \alpha + n, 1 + \beta + n)}{(1 + \alpha + \beta + n, n)} (s - r)^{1+\alpha+\beta+2n}. \tag{2}$$

If f is holomorphic on a neighborhood of the line segment $[r, s]$, then

$$\int_r^s f(x)p_n(x)(x - r)^\beta (s - x)^\alpha \, dx = \frac{1}{n!} I_n F^{(n)}(1 + \alpha + n, 1 + \beta + n; r, s), \tag{3}$$

where $\mathrm{ph}(x - r) = \mathrm{ph}(s - r) = \mathrm{ph}(s - x)$.

Proof Denote the left side of (3) by A_n and let $g(x) = (x - r)^{\beta+n}(s - x)^{\alpha+n}$. The Remark following (1) shows that

$$(1 + \alpha + \beta + n, n)A_n = (-1)^n \int_r^s f(x)g^{(n)}(x)\, dx.$$

If $n \geq 1$, integration by parts changes the right side to

$$(-1)^n[f(x)g^{(n-1)}(x)]_{x=r}^{x=s} + (-1)^{n-1} \int_r^s f^{(1)}(x)g^{(n-1)}(x)\, dx.$$

From Leibnitz' formula for differentiation of a product it follows that

$$g^{(n-1)}(x) = (x - r)^{\beta+1}(s - x)^{\alpha+1}h(x),$$

where h is a polynomial. Since $\alpha + 1$, $\beta + 1 \in \mathbb{C}_>$ and f is continuous on $[r, s]$, the boundary term vanishes. If $n \geq 2$, we integrate by parts again. This time the boundary term contains a factor $(x - r)^{\beta+2}(s - x)^{\alpha+2}$ and again vanishes. After n integrations by parts we have

$$(1 + \alpha + \beta + n, n)A_n = \int_r^s f^{(n)}(x)g(x)\, dx.$$

The integral can now be evaluated as a Dirichlet average of $f^{(n)}$ by using Exercise 5.1-1. ▯

THEOREM 7.8-3 (*Orthogonality of Jacobi polynomials*) Let $1 + \alpha$, $1 + \beta \in \mathbb{C}_>$ and $r, s \in \mathbb{C}$, and assume $r \neq s$. Let $m, n \in \mathbb{N}$ and let δ_{mn} be the Kronecker delta. Define p_n by (7.1-11) and I_n by (2). Then

$$\int_r^s p_m(x)p_n(x)(x - r)^\beta (s - x)^\alpha \, dx = \delta_{mn} I_n, \tag{4}$$

where $\mathrm{ph}(x - r) = \mathrm{ph}(s - r) = \mathrm{ph}(s - x)$.

Proof Since (4) is symmetric in m and n, we may suppose $m \le n$. In (3) we set $f(x) = p_m(x)$ and note that $f^{(n)} = \delta_{mn} n!$. □

The orthogonality of Jacobi polynomials on the interfocal line segment is less general and less fundamental than the biorthogonality (7.2-9) of the polynomials and the functions of the second kind on a contour encircling the segment. Biorthogonality still holds if $r = s$ or if $1 + \alpha$ and $1 + \beta$ have negative real parts (provided $\alpha + \beta \neq -2, -3, \ldots$). Moreover, when orthogonality does hold, it can be deduced from biorthogonality in a way which we shall now sketch without giving full details. If $r \neq s$, we cut the plane along the interfocal line segment $[r, s]$ and deform the contour of biorthogonality by Cauchy's integral theorem so that it consists of the upper and lower edges of the cut, joined by two small circles of radius δ about the ends of the cut. If $1 + \alpha$, $1 + \beta \in \mathbb{C}_>$, one can show that the integrals around the two circles tend to zero with δ. Since the integrals along the edges of the cut are taken in opposite directions, their sum is the integral from r to s of the discontinuity of the integrand $p_m q_n$ across the cut. The polynomial p_m is continuous, but q_n is not. Its discontinuity across the cut at a point x different from r and s is

$$\lim_{\varepsilon \to 0}[q_n(x - \varepsilon) - q_n(x + \varepsilon)] = (2\pi i/I_n)(x - r)^\beta (s - x)^\alpha p_n(x), \tag{5}$$

where $\mathrm{ph}(x - r) = \mathrm{ph}(s - r) = \mathrm{ph}(s - x) = \mathrm{ph}\,\varepsilon - \pi/2$ and I_n is defined by (2). [The case $n = 0$ is equivalent to (6.8-14).] From (5) and (7.2-9) we immediately obtain (4).

If $1 + \alpha$, $1 + \beta \in \mathbb{C}_>$ it is not difficult to prove (5) from (5.1-1) with β, β', x, y, $f(t)$, u respectively replaced by $1 + \alpha + n$, $1 + \beta + n$, $y - r$, $y - s$, $(y - t)^{-n-1}$, $(s - t)/(s - r)$. As $y = x \pm \varepsilon$ approaches a point x on the cut, the straight path of integration may be deformed to make a small semicircular detour around x. On taking the difference in brackets on the left side of (5), the straight parts of the paths cancel to leave an integral around two semicircles forming a full circle. This can be evaluated by Cauchy's integral formula and is essentially the derivative in Rodrigues' formula (1), thus proving (5). [Although we need only the case $1 + \alpha$, $1 + \beta \in \mathbb{C}_>$, (5) is actually valid for $(1 + \alpha, 1 + \beta) \in U_2$ by the permanence of functional relations.]

7.9 Laguerre polynomials

The orthogonality of Jacobi polynomials on the interfocal line segment becomes orthogonality on a half-line as one focus tends to infinity. The Jacobi polynomials, which are R polynomials with two arguments and therefore ${}_2F_1$ polynomials, become ${}_2F_0$ polynomials or equivalently ${}_1F_1$

polynomials. Laguerre studied a special case in 1879 and Sonine the general case in 1880, but they were anticipated in part by Murphy (1833). The practical importance of Laguerre (or Sonine) polynomials was enhanced by Schrödinger's wave mechanics, where they occur in the radial wave functions of the hydrogen atom (Example 7.9-6).

The ellipse of convergence of a Jacobi series becomes a parabola of convergence for a series of Laguerre polynomials. This is understandable from the succession of conic sections. A plane perpendicular to the axis of a right circular cone in $\mathbb{R}^3$ cuts the cone in a circle, which becomes an ellipse if the plane is tilted. As the tilt increases and the plane becomes parallel to a generator of the cone, one focus of the ellipse recedes to infinity and the ellipse becomes a parabola. Circle, ellipse, and parabola are the respective boundaries of convergence of Taylor, Jacobi, and Laguerre series. Further tilt yields a hyperbola, but a corresponding polynomial series does not seem to have been identified.

Laguerre polynomials are limits, as one focus recedes to infinity, of Jacobi polynomials but not of Gegenbauer polynomials because one of the parameters α, β tends to infinity in the limiting process. To understand why this is necessary, we put one focus (say r) at 0 and note that the weight function for orthogonality on $[0, s]$ is proportional to $x^\beta(s-x)^\alpha = s^\alpha x^\beta(1-x/s)^\alpha$ by (7.8-4). To keep the weight function integrable with respect to x as $s \to \infty$, we choose $\alpha = s$ and obtain a weight function proportional to $x^\beta e^{-x}$ by (5.10-1). We shall now obtain the monic Laguerre polynomial as the limit of a monic Jacobi polynomial as $\alpha = s \to \infty$, the other focus being $r = 0$.

THEOREM 7.9-1 Let $n \in \mathbb{N}$ and $\beta, x \in \mathbb{C}$. Then

$$\lim_{s\to\infty} R_n(-s-n, -\beta-n; x, x-s) = x^n\, {}_2F_0(-n, -\beta-n; -1/x). \quad (1)$$

Proof By (6.2-3),

$$R_n(-s-n, -\beta-n; x, x-s)$$
$$= \frac{n!}{(-s-\beta-2n, n)} \sum_{m=0}^{n} \frac{(-s-n, m)(-\beta-n, n-m)}{m!\,(n-m)!} x^m(x-s)^{n-m}.$$

If numerator and denominator are divided by $(-s)^n$, all factors involving s on the right side may be replaced by unity as $s \to \infty$. Thus,

$$R_n(-s-n, -\beta-n; x, x-s) \to \sum_{m=0}^{n} \binom{n}{m}(-\beta-n, n-m)x^m$$
$$= x^n \sum_{m=0}^{n} \binom{n}{m}(-\beta-n, m)x^{-m}.$$

By (2.2-12) this is the same as (1). □

The monic Laguerre polynomial is a polynomial of degree n in β as well as in x. In terms of the customary notation $L_n{}^\beta$ it is

$$(-1)^n n!\, L_n{}^\beta(x) = \sum_{m=0}^{n} \binom{n}{m} (-\beta - n, n - m) x^m$$
$$= x^n\, {}_2F_0(-n, -\beta - n; -1/x). \tag{2}$$

By Exercise 2.4-3 this is equivalent to (2.4-18),

$$L_n{}^\beta(x)/L_n{}^\beta(0) = {}_1F_1(-n; 1 + \beta; x) = S(-n, 1 + \beta + n; x, 0),$$
$$L_n{}^\beta(0) = (1 + \beta, n)/n!. \tag{3}$$

Laguerre's special case is $L_n{}^0$, commonly denoted by L_n.

The limit corresponding to (1) for the Jacobi function of the second kind can be determined as follows if $1 + \beta + n \in \mathbb{C}_>$. In the integral representation (7.4-12) replace x^2 and y^2 by x and $x - s$, and set $\alpha = s$ and $u = s/(t + s)$. Take the limit as $s \to \infty$ under the integral sign (see Appendix B.1) to show that

$$\lim_{s \to \infty} R_{-n-1}(1 + s + n, 1 + \beta + n; x, x - s)$$
$$= x^{-n-1}{}_2F_0(1 + n, 1 + \beta + n; 1/x). \tag{4}$$

DEFINITION 7.9-2 Let $n \in \mathbb{N}$ and $\beta, x \in \mathbb{C}$. Define the monic Laguerre polynomial and monic Laguerre function of the second kind by

$$\begin{aligned} \bar{p}_n(x) &= x^n\, {}_2F_0(-n, -\beta - n; -1/x), \\ \bar{q}_n(x) &= x^{-n-1}\, {}_2F_0(1 + n, 1 + \beta + n; 1/x), \qquad x \in \mathbb{C} - \mathbb{R}_+. \end{aligned} \tag{5}$$

Owing to (7.2-9) one suspects that the $\bar{p}$'s and $\bar{q}$'s are biorthogonal (in some sense which makes the integrals convergent) on a parabola with focus 0 and the positive real line as axis, but this question has apparently never been studied. If f is holomorphic inside such a parabola and satisfies a suitable growth condition, it can be expanded [see Szász (1958)] in a series of Laguerre polynomials with $1 + \beta \in \mathbb{R}_>$,

$$f(x) = \sum_{n=0}^{\infty} a_n \bar{p}_n(x). \tag{6}$$

In view of Theorems 7.6-2 and 7.9-1 it would not be surprising if this remained true for all complex β. We shall assume that it is true for $1 + \beta \in \mathbb{C}_>$ and that term-by-term integration is permissible. Since biorthogonality has not been established, we shall use orthogonality to determine the coefficients a_n.

The first step is to find a Rodrigues' formula for $\bar{p}_n$. By (7.8-1),

$$\begin{aligned}\bar{p}_n(x) &= \lim_{s\to\infty} R_n(-s-n, -\beta-n; x, x-s)\\ &= (-1)^n x^{-\beta} \lim_{s\to\infty} \frac{(s-x)^{-s}}{(1+s+\beta+n, n)} D^n[x^{\beta+n}(s-x)^{s+n}]\\ &= (-1)^n x^{-\beta} \lim_{s\to\infty} \left(1-\frac{x}{s}\right)^{-s} D^n\left[x^{\beta+n}\left(1-\frac{x}{s}\right)^{s+n}\right]. \end{aligned} \tag{7}$$

By the limit formula (5.10-1) for the exponential function (7) suggests that

$$\bar{p}_n(x) = (-1)^n x^{-\beta} e^x D^n(x^{\beta+n}e^{-x}), \tag{8}$$

a result which can be verified directly by Leibnitz' rule with less trouble than it takes to justify interchanging limit and differentiation in (7). We shall now obtain the analog of (7.8-3) by using the Euler measure (3.11-1).

THEOREM 7.9-3 Let $n \in \mathbb{N}$, $1+\beta \in \mathbb{C}_>$, and $x \in \mathbb{R}_+$. Define the measure $\lambda_{1+\beta}$ on $\mathbb{R}_+$ by $d\lambda_{1+\beta}(x) = [\Gamma(1+\beta)]^{-1}x^{\beta}e^{-x}\,dx$. Define $\bar{p}_n$ by (5) and let

$$\bar{I}_n = n!\,(1+\beta, n). \tag{9}$$

Let f be holomorphic on a neighborhood of the positive real line and assume $f^{(m)}(x)e^{-\varepsilon x} \to 0$ as $x \to +\infty$ for some $\varepsilon < 1$ and for $0 \le m \le n-1$. Then

$$\int_0^\infty f(x)\bar{p}_n(x)\,d\lambda_{1+\beta}(x) = \frac{1}{n!}\bar{I}_n \int_0^\infty f^{(n)}(x)\,d\lambda_{1+\beta+n}(x). \tag{10}$$

Proof Let $g(x) = x^{\beta+n}e^{-x}$. By (8),

$$\bar{p}_n(x)\,d\lambda_{1+\beta}(x) = (-1)^n[\Gamma(1+\beta)]^{-1}g^{(n)}(x)\,dx.$$

Substituting in the left side of (10) and integrating by parts n times, we find that the boundary terms vanish each time. The integral becomes

$$[\Gamma(1+\beta)]^{-1}\int_0^\infty f^{(n)}(x)g(x)\,dx,$$

which equals the right side of (10). □

THEOREM 7.9-4 (*Orthogonality of Laguerre polynomials*) Let $m, n \in \mathbb{N}$ and $1+\beta \in \mathbb{C}_>$. Define $\bar{p}_n$ by (5). Then

$$\int_0^\infty \bar{p}_m(x)\bar{p}_n(x)\,d\lambda_{1+\beta}(x) = \bar{I}_n\,\delta_{mn}. \tag{11}$$

Proof Since the integral is symmetric in m and n, we may assume $m \le n$. Setting $f(x) = \bar{p}_m(x)$ in (10), we note that $f^{(n)} = n!\,\delta_{mn}$ and $\lambda_{1+\beta+n}(\mathbb{R}_+) = 1$. □

Finally we multiply (6) by $\bar{p}_m(x)$ and integrate term by term with respect to $\lambda_{1+\beta}$ to obtain

$$l_m a_m = \int_0^\infty f(x)\bar{p}_m(x)\, d\lambda_{1+\beta}(x). \tag{12}$$

By (10),

$$a_n = \frac{1}{n!}\int_0^\infty f^{(n)}(x)\, d\lambda_{1+\beta+n}(x).$$

This limiting case of (7.6-8) shows that the Laguerre series coefficient a_n is an average of the Taylor series coefficient over the positive real line with respect to the measure $\lambda_{1+\beta+n}$.

Laguerre polynomials occur in eigenvalue problems of wave mechanics as the exceptional cases of the following theorem.

THEOREM 7.9-5 Let $x \in \mathbb{R}_+$ and $\beta,\ \beta' \in \mathbb{C}$. Then $|S(\beta,\ \beta';\ x,\ -x)/\Gamma(\beta + \beta')| \to \infty$ as $x \to \infty$ unless $-\beta \in \mathbb{N}$.

Proof Put $y = -x$ in (5.12-18), which holds for $\beta,\ \beta' \in \mathbb{C}$ by Corollary 6.3-3 and the permanence of functional relations. The first term tends to ∞ with x unless $1/\Gamma(\beta) = 0$. □

EXAMPLE 7.9-6 (*Hydrogen atom*) By (5.8-22) the regular Coulomb wave function of a particle with radial coordinate r is

$$r^l S(1 + l + i\eta,\ 1 + l - i\eta;\ -ikr,\ ikr), \tag{13}$$

where $l \in \mathbb{N}$ is the quantum number of angular momentum, k the wave number, and ηk a real constant which is negative if the Coulomb force is attractive. In unbound scattering states k is real, but in bound atomic states it is purely imaginary, and therefore we set $k = i\kappa$ and $\eta = iv$, where κ and v are chosen to be positive. In the case of hydrogen $1/\kappa v$ is a universal constant a called the first Bohr radius. The wave function,

$$r^l S(1 + l - v,\ 1 + l + v;\ r/va,\ -r/va), \tag{14}$$

must tend to 0 as $r \to \infty$ in order to describe a bound state. By Theorem 7.9-5 this requires $1 + l - v = -n$, where $n \in \mathbb{N}$, and consequently $v = N$, where $N = 1 + l + n$ is a positive integer. By (3) a radial wave function of the hydrogen atom is essentially a Laguerre polynomial,

$$\begin{gathered} r^l S(-n,\ 2 + 2l + n;\ r/Na,\ -r/Na) = \frac{n!}{(2 + 2l,\ n)} r^l e^{-r/Na} L_n^{2l+1}(2r/Na), \\ l,\ n \in \mathbb{N}, \qquad N = 1 + l + n. \end{gathered} \tag{15}$$

[Choosing κ and ν negative would have led to the same result, since (14) is even in ν.] Since $\kappa = 1/Na$ and the energy is $E = -\frac{1}{2}e^2 a\kappa^2$, where e is the charge of the electron, the state (15) has energy $E = -e^2/2aN^2$. Each positive integer N corresponds to a discrete energy level of the hydrogen atom, but states (15) with different values of l and n may have the same energy.

EXAMPLE 7.9-7 (*Isotropic harmonic oscillator*) A particle in $\mathbb{R}^q$ which is attracted to the origin by a restoring force proportional to its radial coordinate r is governed by the Schrödinger equation,

$$(\nabla^2 + k^2 - r^2/a^4)\psi = 0. \tag{16}$$

The wave number k and the characteristic length a (called the amplitude of zero-point vibration) are related to the energy E by

$$E = \tfrac{1}{2}\varepsilon k^2 a^2, \tag{17}$$

where the characteristic energy ε will prove to be the spacing between adjacent energy levels. By separating angular coordinates from the Laplacian operator ∇^2 in $\mathbb{R}^q$, $q \geq 2$ [see Hochstadt (1971, Chapter 8, Section 1) and Bateman (1932, Section 6.51)], we obtain the radial wave equation,

$$\frac{d^2\varphi}{dr^2} + \frac{q-1}{r}\frac{d\varphi}{dr} + \left[k^2 - \frac{r^2}{a^4} - \frac{l(l+q-2)}{r^2}\right]\varphi = 0, \tag{18}$$

where $l \in \mathbb{N}$ is the quantum number of angular momentum. If $q = 1$, comparison with (16) shows that (18) still holds provided $l = 0$ or 1. (We shall return to this case in Example 7.10-1 and find that these two values correspond to wave functions φ which are respectively even or odd in $x = r$.)

In (5.8-20) and (5.8-21) we set $y = r^2$ and $\omega(y) = \varphi(r)$, choose the upper signs, and replace ik by $1/2a^2$ to obtain

$$\varphi = r^{2\mu-2\lambda}S(\tfrac{1}{2} + \mu + \nu, \tfrac{1}{2} + \mu - \nu; r^2/2a^2, -r^2/2a^2), \tag{19}$$

$$\frac{d^2\varphi}{dr^2} + \frac{4\lambda+1}{r}\frac{d\varphi}{dr} + \left(\frac{-4\nu}{a^2} - \frac{r^2}{a^4} + \frac{4\lambda^2 - 4\mu^2}{r^2}\right)\varphi = 0. \tag{20}$$

Since (20) coincides with (18) if $4\lambda = q - 2$, $4\nu = -k^2a^2$, and $4\mu = 2l + q - 2$, substitution of these values in (19) yields the regular solution of (18),

$$\varphi = r^l S[\tfrac{1}{4}(2l + q - k^2a^2), \tfrac{1}{4}(2l + q + k^2a^2); r^2/2a^2, -r^2/2a^2]. \tag{21}$$

Acceptable wave functions must tend to 0 as $r \to \infty$. By Theorem 7.9-5 this requires $2l + q - k^2a^2 = -4n$, where $n \in \mathbb{N}$. Thus, by (3), the radial wave functions are essentially Laguerre polynomials,

$$\varphi = r^l S(-n, l + \tfrac{1}{2}q + n; r^2/2a^2, -r^2/2a^2)$$
$$= \frac{n!}{(l + \frac{1}{2}q, n)} r^l e^{-r^2/2a^2} L_n^{l-1+q/2}(r^2/a^2). \tag{22}$$

By (17) the corresponding energy is $E = (2n + l + \frac{1}{2}q)\varepsilon$.

7.10 Hermite polynomials

The orthogonality of Jacobi polynomials on the interfocal line segment becomes orthogonality on a line as the foci tend to infinity in opposite directions. The Jacobi polynomials become ${}_2F_0$ polynomials of a type which Laplace introduced into probability theory in 1812 and Hermite extended to several variables in 1864. In wave mechanics the Hermite polynomials in one variable occur in the wave functions of the one-dimensional harmonic oscillator (see Example 7.10-5).

In the limit the elliptic disk of convergence of a Jacobi series becomes a strip of convergence for a series of Hermite polynomials. An ellipse in the x plane with foci $\pm s$ which passes through the point y has the equation

$$|x + s| + |x - s| = |y + s| + |y - s|. \tag{1}$$

As $s \to \infty$ through real positive values,

$$|s + x| = [(s + \operatorname{Re} x)^2 + (\operatorname{Im} x)^2]^{1/2}$$
$$= s + \operatorname{Re} x + (\operatorname{Im} x)^2/2s + O(1/s^2),$$
$$|s + x| + |s - x| = 2s + (\operatorname{Im} x)^2/s + O(1/s^2).$$

Thus (1) becomes $(\operatorname{Im} x)^2 = (\operatorname{Im} y)^2$. If $\operatorname{Im} y \neq 0$, this is the equation of a pair of parallel lines, one of them passing through y. They form the boundary of a strip having the real axis as center line. Circle, ellipse, and parallel lines (the respective boundaries of convergence of Taylor, Jacobi, and Hermite series) succeed one another as intersections of a plane and a circular cylinder when the tilt of the plane increases.

Hermite polynomials are limits of Gegenbauer polynomials, because both foci tend to infinity and can be treated symmetrically by choosing $\alpha = \beta$ and $r = -s$. The weight function for orthogonality on $[-s, s]$ is then proportional to

$$(x + s)^\alpha (s - x)^\alpha = s^{2\alpha}(1 - x^2/s^2)^\alpha.$$

To keep the weight function integrable with respect to x as $s \to \infty$, we set $\alpha = s^2$ and obtain a weight function proportional to $\exp(-x^2)$ by (5.10-1). We shall now obtain the monic Hermite polynomial as the limit of a monic Gegenbauer polynomial as $\alpha = \beta = s^2 \to \infty$, the foci being $-s$ and s.

THEOREM 7.10-1 Let $n \in \mathbb{N}$ and $x \in \mathbb{C}$. Then

$$\lim_{s \to \infty} R_n(-s^2 - n, -s^2 - n; x + s, x - s) = x^n \, {}_2F_0\left(\frac{-n}{2}, \frac{1-n}{2}; -\frac{1}{x^2}\right). \tag{2}$$

Proof If $a \in \mathbb{R}$, let $[a]$ denote the largest integer not exceeding a. By (6.4-2) and Theorem 6.9-1,

$$\begin{aligned} &R_n(-s^2 - n, -s^2 - n; x + s, x - s) \\ &\quad = \sum_{m=0}^{n} \binom{n}{m} x^{n-m} R_m(-s^2 - n, -s^2 - n; s, -s) \\ &\quad = \sum_{k=0}^{[n/2]} \binom{n}{2k} x^{n-2k} \frac{(\frac{1}{2}, k)}{(-s^2 - n + \frac{1}{2}, k)} s^{2k} \\ &\quad \to x^n \sum_{k=0}^{[n/2]} \frac{(-n, 2k)(\frac{1}{2}, k)}{(2k)!} \left(-\frac{1}{x^2}\right)^k, \qquad s \to \infty. \end{aligned}$$

According to (2.2-7) the last expression equals the right side of (2). ☐

Two notations are in common use: The right side of (2) is denoted by $2^{-n}H_n(x)$, while $He_n(x)$ designates the monic polynomial obtained by setting $-2/x^2$ in place of $-1/x^2$ as the argument of ${}_2F_0$. The notation H_n is customary in wave mechanics, but He_n is preferable in statistics because the weight function $\exp(-x^2/2)$ which goes with $He_n(x)$ corresponds to a normal distribution with unit variance. Hermite polynomials of degree $2m$ and $2m + 1$ are expressible by Exercise 2.4-3 in terms of the Laguerre polynomials $L_m^{-1/2}$ and $L_m^{1/2}$, respectively.

The limit of the Jacobi function of the second kind with $\alpha = \beta = s^2$ is determined by applying the transformation (6.9-7) before using the integral representation (5.9-1) and letting $s \to \infty$. The result is

$$\begin{aligned} &\lim_{s \to \infty} R_{-n-1}(1 + s^2 + n, 1 + s^2 + n; x + s, x - s) \\ &\quad = x^{-n-1} \, {}_2F_0\left(1 + \frac{n}{2}, \frac{1+n}{2}; \frac{1}{x^2}\right). \end{aligned} \tag{3}$$

DEFINITION 7.10-2 Let $n \in \mathbb{N}$ and $x \in \mathbb{C}$. Define the monic Hermite polynomial and monic Hermite function of the second kind by

$$\begin{aligned} \tilde{p}_n(x) &= x^n \, {}_2F_0\left(\frac{-n}{2}, \frac{1-n}{2}; -\frac{1}{x^2}\right), \\ \tilde{q}_n(x) &= x^{-n-1} \, {}_2F_0\left(1+\frac{n}{2}, \frac{1+n}{2}; \frac{1}{x^2}\right), \qquad x \in \mathbb{C} - \mathbb{R}. \end{aligned} \tag{4}$$

The $\tilde{p}$'s and $\tilde{q}$'s are related to special cases of the parabolic cylinder functions, n being an unrestricted complex number in the general case.

If f is holomorphic on a strip having the real axis as center line and if f satisfies a suitable growth condition [Hille (1940)], it can be expanded in a series of Hermite polynomials,

$$f(x) = \sum_{n=0}^{\infty} a_n \tilde{p}_n(x). \tag{5}$$

To determine the coefficients, we first seek a Rodrigues formula for $\tilde{p}_n$. By (7.8-1),

$$\begin{aligned} \tilde{p}_n(x) &= \lim_{s\to\infty} R_n(-s^2-n, -s^2-n; x+s, x-s) \\ &= (-1)^n \lim_{s\to\infty} \frac{(s^2-x^2)^{-s^2}}{(1+2s^2+n, n)} D^n(s^2-x^2)^{s^2+n} \\ &= \left(-\frac{1}{2}\right)^n \lim_{s\to\infty} \left(1-\frac{x^2}{s^2}\right)^{-s^2} D^n\left(1-\frac{x^2}{s^2}\right)^{s^2+n}. \end{aligned}$$

By the limit formula (5.10-1) the preceding equation suggests that

$$\tilde{p}_n(x) = (-\tfrac{1}{2})^n e^{x^2} D^n e^{-x^2}, \tag{6}$$

a result that can be verified directly.

We shall now deduce the analog of (7.8-3) and (7.9-10). We define a measure γ on $\mathbb{R}$, satisfying $\gamma(\mathbb{R}) = 1$, by

$$d\gamma(x) = \pi^{-1/2} e^{-x^2}\, dx. \tag{7}$$

THEOREM 7.10-3 Let $n \in \mathbb{N}$ and $x \in \mathbb{R}$. Define $\tilde{p}_n$ by (4) and let

$$\tilde{I}_n = n!\,2^{-n} \tag{8}$$

Let f be holomorphic on a neighborhood of the real line and assume that $f^{(m)}(x)\exp(-\varepsilon x^2) \to 0$ as $|x| \to \infty$ for some $\varepsilon < 1$ and for $0 \le m \le n-1$. Then

$$\int_{-\infty}^{\infty} f(x)\tilde{p}_n(x)\, d\gamma(x) = \frac{1}{n!} \tilde{I}_n \int_{-\infty}^{\infty} f^{(n)}(x)\, d\gamma(x). \tag{9}$$

Proof Substitute (6) and integrate by parts n times. The boundary terms vanish each time because of the condition imposed on $f^{(m)}$. □

THEOREM 7.10-4 (*Orthogonality of Hermite polynomials*) Let $m, n \in \mathbb{N}$ and define $\tilde{p}_n$ by (4). Then

$$\int_{-\infty}^{\infty} \tilde{p}_m(x)\tilde{p}_n(x)\, d\gamma(x) = \tilde{I}_n\, \delta_{mn}. \tag{10}$$

Proof By symmetry in m and n we may assume $m \le n$. Setting $f(x) = p_m(x)$ in (9), we note that $f^{(n)} = n!\, \delta_{mn}$ and $\gamma(\mathbb{R}) = 1$. □

We assume that (5) may be integrated term by term with respect to γ after multiplication by $\tilde{p}_m(x)$. Then the coefficient a_m is given by

$$\tilde{I}_m a_m = \int_{-\infty}^{\infty} f(x)\tilde{p}_m(x)\, d\gamma(x). \tag{11}$$

By (9),

$$a_n = \frac{1}{n!}\int_{-\infty}^{\infty} f^{(n)}(x)\, d\gamma(x). \tag{12}$$

This limiting case of (7.6-8) shows that the Hermite series coefficient is an average of the Taylor series coefficient over the real line with respect to the measure γ.

EXAMPLE 7.10-5 (*Harmonic oscillator*) Although other methods are more direct, it is interesting to see how the wave functions of the one-dimensional harmonic oscillator can be found from (7.9-22) and (7.9-2) by setting $q = 1$ and $r = x$:

$$\begin{aligned}\varphi &= x^l S(-n,\, l + \tfrac{1}{2} + n;\, x^2/2a^2,\, -x^2/2a^2)\\ &= \text{const}\; x^{2n+l} e^{-x^2/2a^2}\, {}_2F_0(-n, \tfrac{1}{2} - l - n;\, -a^2/x^2),\end{aligned} \tag{13}$$

where $n \in \mathbb{N}$ and $l = 0$ or 1. Let us define $m = 2n + l$. According as $l = 0$ or 1, it follows that m is even or odd, $n = \frac{1}{2}m$ or $\frac{1}{2}(m-1)$, and $n + l - \frac{1}{2} = \frac{1}{2}(m-1)$ or $\frac{1}{2}m$. Aside from a multiplicative constant, (13) becomes

$$\begin{aligned}&(x/a)^m e^{-x^2/2a^2}\, {}_2F_0\left(\frac{-m}{2}, \frac{1-m}{2};\, -\frac{a^2}{x^2}\right)\\ &\qquad = e^{-x^2/2a^2}\tilde{p}_m(x/a) = 2^{-m} e^{-x^2/2a^2} H_m(x/a),\end{aligned} \tag{14}$$

where $m \in \mathbb{N}$ and H_m is the usual symbol defined after Theorem 7.10-1. By the last sentence of Example 7.9-7 the corresponding energy is $E = (m + \frac{1}{2})\varepsilon$.

NOTES

An elementary discussion of Jacobi polynomials is given by Jackson (1941), and some formal properties and generating functions are derived by Rainville (1960, Chapter 16). The classic treatise on the subject is Szegö (1975); a shorter account is found in Erdélyi (1953, Chapter X), and recent developments are surveyed by Askey (1975), particularly those connected with positivity. For the history of addition formulas see the Notes at the end of Chapter 1. It was first observed in Boas (1964) that orthogonality is not essential to the expansion of an analytic function in Jacobi series. The biorthogonality property was noticed by Colton (1968). For a different proof of Lemma 7.6-1 see Carlson (1974b).

Bateman (1932, pp. 451–454) gives references to some early work on Laguerre polynomials and Hermite polynomials.

EXERCISES

7.1-1 Let $n \in \mathbb{N}$, $(1+\alpha, 1+\beta) \in U^2 \cap U_2$, and $x, y \in \mathbb{C}$. Show that

$$R_n(-\alpha-n, -\beta-n; x, y) = \frac{n!(1+\alpha, n)(1+\beta, n)}{(1+\alpha+\beta+n, n)} \sum_{m=0}^{n} \frac{x^m y^{n-m}}{(1+\beta, m)(1+\alpha, n-m)m!(n-m)!}.$$

7.1-2 Let $t, x, y \in \mathbb{C}$ and $(1+\alpha, 1+\beta) \in U^2$. Prove Bateman's generating relation,

$${}_0F_1(1+\beta, tx)\,{}_0F_1(1+\alpha, ty) = \sum_{n=0}^{\infty} t^n \frac{(1+\alpha+\beta+n, n)}{(1+\alpha, n)(1+\beta, n)n!} R_n(-\alpha-n, -\beta-n; x, y).$$

Because R_n is homogeneous, there is no loss of generality in setting $t = 1$.

7.1-3 Jacobi polynomials are often defined by

$$P_n^{(\alpha,\beta)}(x) = 2^{-n} \sum_{m=0}^{n} \binom{\alpha+n}{m}\binom{\beta+n}{n-m}(x+1)^m(x-1)^{n-m}.$$

Show that

$$P_n^{(\alpha,\beta)}(x) = \frac{(1+\alpha+\beta+n, n)}{2^n n!} R_n(-\alpha-n, -\beta-n; x+1, x-1),$$

and that $P_n^{(\alpha,\beta)}(x)$ can be written as a ${}_2F_1$ polynomial with any one of the arguments

$$\frac{2}{1-x},\ \frac{2}{1+x},\ \frac{1-x}{2},\ \frac{1+x}{2},\ \frac{x+1}{x-1},\ \frac{x-1}{x+1}.$$

The first few polynomials are

$$P_0^{(\alpha,\beta)} = 1; \qquad 2P_1^{(\alpha,\beta)}(x) = (\alpha+\beta+2)x + \alpha - \beta;$$

$$8P_2^{(\alpha,\beta)}(x) = (\alpha+\beta+3)(\alpha+\beta+4)x^2 + 2(\alpha-\beta)(\alpha+\beta+3)x - (\alpha+\beta+4) + (\alpha-\beta)^2.$$

The Legendre polynomial is $P_n = P_n^{(0,0)}$.

7.1-4 Jacobi functions of the second kind are often defined by

$$Q_n^{(\alpha,\beta)}(x) = 2^{\alpha+\beta+n}B(1+\alpha+n, 1+\beta+n)(x-1)^{-\alpha-n-1}(x+1)^{-\beta}$$

$$\cdot\, {}_2F_1\left(1+n, 1+\alpha+n; 2+\alpha+\beta+2n; \frac{2}{1-x}\right),$$

where $n \in \mathbb{N}$, $(1+\alpha, 1+\beta) \in U^2$, and $x \in \mathbb{C} - [-1, 1]$. The Legendre function of the second kind is $Q_n(x) = Q_n^{(0,0)}(x)$. Show that

$$\begin{aligned} &Q_n^{(\alpha,\beta)}(x) \\ &= 2^{\alpha+\beta+n}\mathrm{B}(1+\alpha+n, 1+\beta+n)(x+1)^{-\beta}(x-1)^{-\alpha}R_{-n-1}(1+\alpha+n, 1+\beta+n; x+1, x-1) \\ &= 2^{-t-1}\mathrm{B}(-\alpha-t, -\beta-t)R_t(-\alpha-t, -\beta-t; x+1, x-1), \qquad -t = 1+\alpha+\beta+n. \end{aligned}$$

7.1-5 Show that the coefficients a_{nm} and b_{nm} in

$$p_n(x) = \sum_{m=0}^{n} a_{nm}x^m, \qquad q_n(y) = \sum_{m=0}^{\infty} b_{nm}y^{-n-1-m}, \qquad |y| > |r|, \quad |y| > |s|,$$

are given by

$$a_{nm} = \binom{n}{m}R_{n-m}(-\alpha-n, -\beta-n; -r, -s), \qquad b_{nm} = \binom{n+m}{m}R_m(1+\alpha+n, 1+\beta+n; r, s).$$

7.1-6 If $n-1 \in \mathbb{N}$ and $(1+\alpha, 1+\beta) \in U_2$, show that

$$p_{n+1}(x) + (V_n - x)p_n(x) + W_n p_{n-1}(x) = 0, \qquad q_{n-1}(y) + (V_n - y)q_n(y) + W_{n+1}q_{n+1}(y) = 0,$$

where

$$V_n = \frac{(\alpha^2-\beta^2)(r-s)}{2(\alpha+\beta+2n)(\alpha+\beta+2n+2)} + \frac{r+s}{2}, \qquad W_n = \frac{n(\alpha+n)(\beta+n)(\alpha+\beta+n)(r-s)^2}{(\alpha+\beta+2n)^2[(\alpha+\beta+2n)^2-1]}.$$

Removable singularities of V_n and W_n are to be removed by requiring continuity. The recurrence relations hold also for $n = 0$ if we define $q_{-1} = 1$ and $W_0 p_{-1} = 0$.

7.1-7 If $N \in \mathbb{N}$, show that

$$1 = (y-x)\sum_{n=0}^{N} p_n(x)q_n(y) + p_{N+1}(x)q_N(y) - W_{N+1}p_N(x)q_{N+1}(y),$$

where W_{N+1} is defined in Exercise 7.1-6. This relation is sometimes called Christoffel's second summation formula.

7.1-8 Let a prime denote differentiation with respect to x. For every $n \in \mathbb{N}$ show that

$$(x-r)(x-s)p_n'' + [(1+\alpha)(x-r) + (1+\beta)(x-s)]p_n' - n(1+\alpha+\beta+n)p_n = 0,$$

$$(x-r)(x-s)q_n'' + [(1-\alpha)(x-r) + (1-\beta)(x-s)]q_n' - (n+1)(\alpha+\beta+n)q_n = 0.$$

Show that these two equations are adjoint differential equations. Show also that $(x-r)^{-\beta}(x-s)^{-\alpha}q_n$ satisfies the same differential equation as p_n. This explains the choice of $Q_n^{(\alpha,\beta)}$ in Exercise 7.1-4. Note that $Q_n^{(\alpha,\beta)}$ has an extra branch point at infinity.

7.1-9 Let $x, y \in \mathbb{C}$ and $\delta \in U$. Show that

$${}_0F_1(\delta; x^2)\,{}_0F_1(\delta; y^2) = \sum_{n=0}^{\infty} \frac{1}{(\delta, n)n!} R_n\left[\delta - \frac{1}{2}, \delta - \frac{1}{2}; (x+y)^2, (x-y)^2\right].$$

This Dirichlet average of a ${}_0F_1$ series is used in Example 7.7-4.

7.1-10 Let $t, \theta \in \mathbb{C}$ and let J_0, I_0, and P_n denote respectively a Bessel function, modified Bessel function, and Legendre polynomial. Show that

$$J_0(2t\sin\theta)I_0(2t\cos\theta) = \sum_{n=0}^{\infty} \frac{t^{2n}}{n!\,n!} P_n(\cos 2\theta).$$

7.1-11 If $n - 1 \in N$, $(1 + \alpha, 1 + \beta) \in \bar{U}_2$, and W_n is defined as in Exercise 7.1-6, show that

$$p_n(x)q_{n-1}(x) - W_n p_{n-1}(x)q_n(x) = 1.$$

7.1-12 Let $n \in \mathbb{N}$ and $\nu, \theta \in \mathbb{C}$. Show that

$$\frac{2^n(\nu, 2n)}{(2n)!} R_n\left(\frac{1}{2} - \nu - n, \frac{1}{2} - n; \cos\theta + 1, \cos\theta - 1\right) = C_{2n}^{\nu}\left(\cos\frac{1}{2}\theta\right),$$

$$\frac{2^{n+1}(\nu, 2n+1)}{(2n+1)!}\left[\cos\left(\frac{1}{2}\theta\right)R_n\left(\frac{1}{2} - \nu - n, -\frac{1}{2} - n; \cos\theta + 1, \cos\theta - 1\right)\right] = C_{2n+1}^{\nu}\left(\cos\frac{1}{2}\theta\right).$$

7.1-13 Let $\alpha, \beta, x, y \in \mathbb{C}$ and $\gamma, \delta \in U$. Assume $|x|^{1/2} + |y|^{1/2} < 1$. Appell's double hypergeometric series F_4 is defined by

$$F_4(\alpha, \beta; \gamma, \delta; x, y) = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \frac{(\alpha, m+n)(\beta, m+n)}{(\gamma, m)(\delta, n)m!\,n!} x^m y^n.$$

The double series converges absolutely [see Bailey (1935, p. 75)]. Show that F_4 is a series of Jacobi polynomials,

$$F_4(\alpha, \beta; \gamma, \delta; x, y) = \sum_{n=0}^{\infty} \frac{(\alpha, n)(\beta, n)(\gamma + \delta - 1 + n, n)}{(\gamma, n)(\delta, n)n!} R_n(1 - \delta - n, 1 - \gamma - n; x, y),$$

and that the case $\gamma = \delta$ is a Dirichlet average of a ${}_2F_1$ series,

$$F_4(\alpha, \beta; \gamma, \gamma; x^2, y^2) = \sum_{n=0}^{\infty} \frac{(\alpha, n)(\beta, n)}{(\gamma, n)n!} R_n\left[\gamma - \frac{1}{2}, \gamma - \frac{1}{2}; (x+y)^2, (x-y)^2\right].$$

7.2-1 Let $m, n \in \mathbb{N}$ and $\nu + \frac{1}{2} \in U$. Let γ be a simple closed contour encircling the line segment $[-1, 1]$ in the positive direction. Show that the Gegenbauer polynomial $C_m^{\nu-1/2}$ satisfies

$$\frac{1}{2\pi i}\int_{\gamma} C_m^{\nu-1/2}(y)R_{-n-1}(\nu + n, \nu + n; y + 1, y - 1)\,dy = \delta_{mn}\frac{2^n(\nu - \frac{1}{2}, n)}{n!}.$$

7.2-2 Let $n \in N$, $(1 + \alpha, 1 + \beta) \in U_2$, and $A, B, x, r, s \in \mathbb{C}$. If p_m is defined by (7.1-11), show that

$$(A + Bx)^n = \sum_{m=0}^{n}\binom{n}{m}B^m R_{n-m}(1 + \alpha + m, 1 + \beta + m; A + Br, A + Bs)p_m(x).$$

7.3-1 Let $n \in \mathbb{N}$ and $x \in \mathbb{C}$. Prove Unsöld's theorem,

$$\sum_{m=-n}^{n}(-1)^m P_n^{\,m}(x)P_n^{-m}(x) = 1.$$

7.3-2 Let **r** and **r**′ be vectors in $\mathbb{R}^3$ with spherical polar coordinates (r, θ, φ) and (r', θ', φ'), respectively. Assume $r \neq r'$, and let $r_< = \min\{r, r'\}$ and $r_> = \max\{r, r'\}$. Show that

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \sum_{n=0}^{\infty}\frac{r_<^{\,n}}{r_>^{\,n+1}}\sum_{m=-n}^{n}(-1)^m P_n^{\,m}(\cos\theta)P_n^{-m}(\cos\theta')e^{im(\varphi-\varphi')}.$$

This expansion is useful in many problems connected with gravitational or Coulomb forces.

7.4-1 Let $n \in \mathbb{N}$ and $x, y \in \mathbb{C}$. Define W by (7.4-2). Show that

$$R_n\left(-\frac{1}{2}-n, -\frac{1}{2}-n; x^2, y^2\right) = \frac{1}{xy}\left[\left(\frac{x+y}{2}\right)^{2n+2} - \left(\frac{x-y}{2}\right)^{2n+2}\right], \qquad xy \neq 0,$$

$$R_{-n-1}\left(\frac{3}{2}+n, \frac{3}{2}+n; x^2, y^2\right) = \left(\frac{x+y}{2}\right)^{-2n-2}, \qquad (x, y) \in W.$$

7.4-2 Let $v, x, y, \theta \in \mathbb{C}$. Assume $x \neq y$, $xy \neq 0$, and $\sin\theta \neq 0$. As $n \to \infty$, show that

$$n^v R_n(v, v; x, y) \sim A_1 x^n + A_2 y^n, \qquad v + \tfrac{1}{2} \in U,$$

$$n^{1-v} C_n^v(\cos\theta) \sim A_1' e^{in\theta} + A_2' e^{-in\theta}, \qquad v \in U.$$

The quantities A_1, A_2, A_1', A_2' are nonzero and independent of n, but A_1' and A_2' may depend on v and θ, and A_1 and A_2 may depend on v, x, y.

7.4-3 Carry farther the saddle-point integrations in the proofs of Theorems 7.4-2 and 7.4-3. Show that

$$C_1 = \left(\frac{x+y}{2x}\right)^{1/2+\beta}\left(\frac{y+x}{2y}\right)^{1/2+\alpha}, \qquad C_2 = \left(\frac{x-y}{2x}\right)^{1/2+\beta}\left(\frac{y-x}{2y}\right)^{1/2+\alpha},$$

$$C_3 = \left(\frac{x+y}{2x}\right)^{1/2-\beta}\left(\frac{y+x}{2y}\right)^{1/2-\alpha}\left(\frac{x+y}{2}\right)^{-2}.$$

7.4-4 Let $n \in \mathbb{N}$ and $v, \theta \in \mathbb{C}$. Assume $\sin\theta \neq 0$. As $n \to \infty$, show that

$$C_n^v(\cos\theta) \sim \frac{(n/2)^{v-1}}{\Gamma(v)\sin^v\theta}\cos[(n+v)\theta - v\pi/2].$$

7.5-1 Let $n \in \mathbb{N}$, $(1+\alpha, 1+\beta) \in U_2$, and $x \in \mathbb{C}$. Show that

$$\limsup_{n\to\infty} |p_n(x)|^{1/n} \leq \mu(x).$$

7.5-2 Let $n \in \mathbb{N}$, $(1+\alpha, 1+\beta) \in U_2$, $r, s \in \mathbb{C}$, and $\rho, M \in \mathbb{R}$. Assume $4\rho > |r-s|$ and define $\hat{q}_n(\rho)$ by (7.5-6). Let f be holomorphic on the open elliptic disk Ω_ρ and assume $|f| \leq M$ on Ω_ρ. Show that

$$|F^{(n)}(1+\alpha+n, 1+\beta+n; r, s)| \leq \frac{4}{\pi} n!\, M\rho\hat{q}_n(\rho).$$

[If $r = s$, then $\hat{q}_n(\rho) = \rho^{-n-1}$ and we recover Cauchy's inequality for $f^{(n)}(r)$ except for the factor $4/\pi$.]

7.6-1 Let γ be a positively oriented rectifiable Jordan curve, and let y be a point in the outer region of γ. Assume that f is holomorphic on γ and the outer region of γ and has a zero at infinity. Show that

$$f(y) = \frac{1}{2\pi i}\int_\gamma \frac{f(w)}{y-w}\,dw.$$

7.6-2 Let $A, B \in \mathbb{C}$ and $1 + 2v \in U$. Let f be holomorphic on an open elliptic disk Ω with foci $A + B$ and $A - B$. If $A + Bx \in \Omega$, show that

$$f(A+Bx) = \sum_{n=0}^{\infty} \frac{(B/2)^n}{(v, n)} F^{(n)}\left(\frac{1}{2}+v+n, \frac{1}{2}+v+n; A+B, A-B\right) C_n^v(x).$$

If $v = 0$, $C_n^v/(v, n)$ has the value $(2-\delta_{n0})T_n/n!$ by (6.10-12) and (6.10-14).

7.6-3 Let $\nu \in \mathbb{C}_>$ and let f be holomorphic on an open elliptic disk Ω with foci -1 and 1. If $n \in \mathbb{N}$, define

$$A_n = \frac{1}{2^n(\nu, n)} F^{(n)}\left(\frac{1}{2} + \nu + n, \frac{1}{2} + \nu + n; -1, 1\right).$$

Let $\theta, \varphi \in \mathbb{C}$ and assume $\cos\theta, \cos(\theta + \varphi), \cos(\theta - \varphi) \in \Omega$. Show that

$$f(\cos\theta) = \sum_{n=0}^{\infty} A_n C_n{}^{\nu}(\cos\theta),$$

$$F[\nu, \nu; \cos(\theta + \varphi), \cos(\theta - \varphi)] = \sum_{n=0}^{\infty} A_n C_n{}^{\nu}(\cos\theta) C_n{}^{\nu}(\cos\varphi)/C_n{}^{\nu}(1).$$

7.6-4 Let $a, b, c \in \mathbb{C}$. Let f be holomorphic on an open elliptic disk with foci $a \pm 2(bc)^{1/2}$ and mean radius $\rho > |bc|^{1/2}$. Let D be an open circular annulus in $\mathbb{C}$ with center 0, inner radius $|c|\rho^{-1}$, and outer radius $\rho|b|^{-1}$. If $z \in D$ prove the Laurent expansion,

$$f(a + bz + cz^{-1}) = F\left[\frac{1}{2}, \frac{1}{2}; a + 2(bc)^{1/2}, a - 2(bc)^{1/2}\right]$$
$$+ \sum_{n=1}^{\infty} \frac{1}{n!} F^{(n)}\left[\frac{1}{2} + n, \frac{1}{2} + n; a + 2(bc)^{1/2}, a - 2(bc)^{1/2}\right](b^n z^n + c^n z^{-n}).$$

7.7-1 Let $\lambda, x \in \mathbb{C}$ and $2\nu \in U$. Prove Sonine's formula,

$$e^{i\lambda x} = 2^{\nu}\Gamma(\nu)\lambda^{-\nu} \sum_{n=0}^{\infty} i^n(\nu + n) J_{\nu+n}(\lambda) C_n{}^{\nu}(x).$$

7.7-2 Let $y \in \mathbb{C}$ and $2\nu \in U$. If $\nu \notin \mathbb{N}$, assume further that $y \neq 0$ and $|\mathrm{ph}\, y| < \pi$. Deduce the Neumann series of a power of y,

$$y^{\nu} = 2^{\nu} \sum_{n=0}^{\infty} \frac{\Gamma(\nu + n)}{n!} (\nu + 2n) J_{\nu+2n}(y).$$

7.7-3 Let $x \in \mathbb{C}$ and $n \in \mathbb{Z}$. Deduce Bessel's representation,

$$2\pi J_n(x) = \int_0^{2\pi} \exp(ix\sin\theta - in\theta)\, d\theta = 2\int_0^{\pi} \cos(x\sin\theta - n\theta)\, d\theta.$$

7.7-4 Let $t \in \mathbb{C}$, $n \in \mathbb{N}$, $k^2 \in \mathbb{C}$, $k^2 \notin [1, +\infty)$. Show that

$$\int_0^{\pi/2} (1 - k^2\sin^2\theta)^t \cos 2n\theta\, d\theta = \frac{\pi}{2}\binom{t}{n}\left(\frac{k^2}{4}\right)^n R_{t-n}\left(\frac{1}{2} + n, \frac{1}{2} + n; 1 - k^2, 1\right).$$

7.7-5 Let $x, y \in \mathbb{C}$ and $2c - 1 \in U$. Show that

$${}_0F_1(c; x + y) = \sum_{n=0}^{\infty} \frac{(-xy)^n}{(c, 2n)(c - 1 + n, n)n!}\, {}_0F_1(c + 2n; x)\, {}_0F_1(c + 2n; y).$$

7.7-6 Let $A, B, \theta \in \mathbb{C}$ and define $w = (A^2 + B^2 - 2AB\cos\theta)^{1/2}$. Show that

$$J_0(w) = J_0(A)J_0(B) + 2\sum_{n=1}^{\infty} J_n(A)J_n(B)\cos n\theta.$$

7.7-7 Let $A, \theta \in \mathbb{C}$ and $2\nu \in U$. Show that

$$\left(2A\sin\frac{1}{2}\theta\right)^{-\nu} J_\nu\left(2A\sin\frac{1}{2}\theta\right) = 2^\nu\Gamma(\nu)\sum_{n=0}^{\infty}(\nu+n)[A^{-\nu}J_{\nu+n}(A)]^2 C_n{}^\nu(\cos\theta).$$

7.7-8 Electric current flows in a circular path, every point of which has radial coordinate a and polar angle α. The magnetic dipole moment of the current loop is m, and its vector potential everywhere points in the azimuthal direction with magnitude

$$A(r,\theta) = \frac{m}{\pi a \sin\alpha}\int_0^{2\pi}[a^2 + r^2 - 2ar(\cos\alpha\cos\theta + \sin\alpha\sin\theta\cos\beta)]^{-1/2}\cos\beta\, d\beta.$$

Let $r_\pm{}^2 = a^2 + r^2 - 2ar\cos(\theta \pm \alpha)$ denote the square of the maximum $(+)$ or minimum $(-)$ distance to the circle from the point where A is evaluated. Show that

$$A(r,\theta) = mr\sin\theta\, R_{-3/2}(\tfrac{3}{2}, \tfrac{3}{2}; r_+{}^2, r_-{}^2).$$

If $r \neq a$, expand A in powers of $t = \min\{a/r, r/a\}$.

7.7-9 Let $\nu \in \mathbb{C}_>$ and $\lambda, \theta, \varphi \in \mathbb{C}$. Assume $|t| < \exp(-|\mathrm{Im}\,\theta| - |\mathrm{Im}\,\varphi|)$. Show that

$$R_{-\lambda}[\nu, \nu; 1 - 2t\cos(\theta+\varphi) + t^2, 1 - 2t\cos(\theta-\varphi) + t^2]$$
$$= \sum_{n=0}^{\infty}\frac{(\lambda, n)}{(\nu, n)} t^n R_{-\lambda-n}\left[\frac{1}{2}+\nu+n, \frac{1}{2}+\nu+n; (1+t)^2, (1-t)^2\right]\frac{C_n{}^\nu(\cos\theta)C_n{}^\nu(\cos\varphi)}{C_n{}^\nu(1)}.$$

Show further that this reduces if $\lambda = \nu$ to (6.11-4) and if $\lambda = \nu + 1$ to the Poisson kernel for Gegenbauer polynomials (see Exercise 7.8-9):

$$(1 - t^2)R_{-\nu-1}[\nu, \nu; 1 - 2t\cos(\theta+\varphi) + t^2, 1 - 2t\cos(\theta-\varphi) + t^2]$$
$$= \sum_{n=0}^{\infty}(1 + n/\nu)t^n C_n{}^\nu(\cos\theta)C_n{}^\nu(\cos\varphi)/C_n{}^\nu(1).$$

7.7-10 Let $\nu \in \mathbb{C}_>$ and $\lambda, \theta, \varphi \in \mathbb{C}$. Show that

$$S[\nu, \nu; \lambda\cos(\theta+\varphi), \lambda\cos(\theta-\varphi)]$$
$$= e^{\lambda\cos\theta\cos\varphi}S(\nu, \nu; \lambda\sin\theta\sin\varphi, -\lambda\sin\theta\sin\varphi)$$
$$= \sum_{n=0}^{\infty}\frac{(\lambda/2)^n}{(\nu, n)} S\left(\frac{1}{2}+\nu+n, \frac{1}{2}+\nu+n; \lambda, -\lambda\right)C_n{}^\nu(\cos\theta)C_n{}^\nu(\cos\varphi)/C_n{}^\nu(1).$$

7.7-11 Let $m \in \mathbb{N}$ and $r, \theta, s, \alpha \in \mathbb{C}$. Define $z = r\cos\theta$, $\rho = r\sin\theta$, $k = s\cos\alpha$, $\gamma = s\sin\alpha$. Show that

$$e^{ikz}J_m(\gamma\rho) = \sum_{n=m}^{\infty} i^{n-m}(2n+1)\frac{(n-m)!}{(n+m)!}j_n(sr)P_n{}^m(\cos\theta)P_n{}^m(\cos\alpha),$$

where J_m, j_n, and $P_n{}^m$ are respectively a Bessel function, spherical Bessel function, and associated Legendre function. This expansion allows a cylindrical wave to be represented as a superposition of spherical waves.

7.7-12 Let $n \in \mathbb{N}$ and $A, B, C \in \mathbb{R}$. Let f be holomorphic on the interval with endpoints $A \pm (B^2 + C^2)^{1/2}$. Show that

$$\int_0^{2\pi} f(A + B\cos\theta + C\sin\theta)e^{\pm in\theta}\,d\theta$$
$$= \frac{2\pi}{n!}\left(\frac{B \pm iC}{2}\right)^n F^{(n)}\left[\frac{1}{2} + n, \frac{1}{2} + n; A + (B^2 + C^2)^{1/2}, A - (B^2 + C^2)^{1/2}\right],$$

where upper (lower) signs go together.

7.7-13 Let $n \in \mathbb{N}$ and $A, B, C \in \mathbb{R}$. Assume $A > (B^2 + C^2)^{1/2}$. Show that

$$\int_0^{2\pi} \frac{e^{\pm in\theta}}{A + B\cos\theta + C\sin\theta}\,d\theta = \frac{2\pi(-1)^n}{(A^2 - B^2 - C^2)^{1/2}}\left[\frac{B \pm iC}{A + (A^2 - B^2 - C^2)^{1/2}}\right]^n,$$

where upper (lower) signs go together.

7.7-14 Let $A, B, x, \lambda, r, s \in \mathbb{C}$ and $(1 + \alpha, 1 + \beta) \in U_2$. Define p_n by (7.11) and μ by (7.5-3). Assume $\mu(x) < \mu(A/B)$. Show that

$$(A - Bx)^{-\lambda} = \sum_{n=0}^{\infty} \frac{(\lambda, n)}{n!} B^n R_{-\lambda-n}(1 + \alpha + n, 1 + \beta + n; A - Br, A - Bs)p_n(x).$$

Note that (7.6-1) and Exercise 7.2-2 are special cases.

7.7-15 Let $\lambda, \nu \in \mathbb{C}$. Let $\mathbf{r}$ and $\mathbf{r}'$ be vectors in $\mathbb{R}^m$, $m \geq 1$, let r and r' denote their lengths, and let θ be the angle between them ($\cos\theta = \mathbf{r} \cdot \mathbf{r}'/rr'$). Assume $r \neq r'$. Show that

$$|\mathbf{r} - \mathbf{r}'|^{-2\lambda} = \sum_{n=0}^{\infty} \frac{(\lambda, n)}{(\nu, n)} (rr')^n R_{-\lambda-n}\left[\frac{1}{2} + \nu + n, \frac{1}{2} + \nu + n; (r + r')^2, (r - r')^2\right] C_n^{\nu}(\cos\theta).$$

If $\nu = 0$, see Exercise 7.6-2. Show that this equation agrees with Exercise 6.7-12.

7.8-1 Let $n \in \mathbb{N}$ and $\nu, x \in \mathbb{C}$ and $D = d/dx$. Show that the Legendre and Gegenbauer polynomials have Rodrigues' formulas,

$$2^n n! P_n(x) = D^n(x^2 - 1)^n,$$
$$2^n n!(\nu + \tfrac{1}{2}, n)C_n^{\nu}(x) = (2\nu, n)(x^2 - 1)^{1/2-\nu} D^n(x^2 - 1)^{n-1/2+\nu}.$$

7.8-2 Let $n, n' \in \mathbb{N}$ and $m \in \mathbb{Z}$. Assume $|m| \leq n$ and $|m| \leq n'$. Show that associated Legendre functions of the same integral order have the orthogonality property

$$\int_{-1}^{1} P_n^m(x)P_{n'}^m(x)\,dx = \delta_{nn'}\frac{2}{2n + 1}\frac{(n + m)!}{(n - m)!}.$$

7.8-3 Let $m, n \in \mathbb{N}$ and $\nu + \frac{1}{2} \in \mathbb{C}_>$. Let $\mu_{(\beta,\beta)}$ be a Dirichlet measure on $[0, 1]$ with parameters $b = (\beta, \beta)$. Show that the Gegenbauer polynomials have the orthogonality property

$$\left[\mathrm{B}\left(\nu + \frac{1}{2}, \frac{1}{2}\right)\right]^{-1} \int_0^{\pi} C_m^{\nu}(\cos\theta)C_n^{\nu}(\cos\theta)(\sin\theta)^{2\nu}\,d\theta$$
$$= \left[\mathrm{B}\left(\nu + \frac{1}{2}, \frac{1}{2}\right)\right]^{-1} \int_{-1}^{1} C_m^{\nu}(x)C_n^{\nu}(x)(1 - x^2)^{\nu-1/2}\,dx$$
$$= \int_0^1 C_m^{\nu}(1 - 2u)C_n^{\nu}(1 - 2u)\,d\mu_{(\nu+1/2,\nu+1/2)}(u) = \partial_{mn}\frac{\nu}{\nu + n}C_n^{\nu}(1).$$

7.8-4 Let $m \in \mathbb{N}$ and $\nu \in \mathbb{C}_>$. Let $\mu_{(\nu,\nu)}$ be a Dirichlet measure on [0, 1]. Show that

$$\int_0^1 C_m^{\nu-1/2}(1-2u)\,d\mu_{(\nu,\nu)}(u) = \delta_{m0},$$

and use this to recover the product formula (6.11-5) from (7.3-8).

7.8-5 Let $n \in \mathbb{N}$ and $\beta, \beta', x, y \in \mathbb{C}$. Show that

$$(\beta+\beta', n)R_n(\beta, \beta'; x, y) = (-1)^n x^{\beta'+n} y^{\beta+n}\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)^n x^{-\beta'} y^{-\beta}.$$

Use this to deduce (7.8-1).

7.8-6 Let $n \in \mathbb{N}$, $b \in \mathbb{C}^k$, and $z \in (\mathbb{C}-\{0\})^k$. Define $c = \sum_{i=1}^k b_i$, $D = \sum_{i=1}^k \partial/\partial z_i$, and $z^{-1} = (z_1^{-1}, \ldots, z_k^{-1})$. Show that

$$(c, n)R_n(b, z^{-1}) = (-1)^n \prod_{i=1}^k z_i^{b_i} D^n \prod_{j=1}^k z_j^{-b_j}.$$

Deduce from this the formula in Exercise 7.8-5.

7.8-7 Let $m \in \mathbb{N}$ and $x \in \mathbb{C}$. Let P_m be a Legendre polynomial and j_m a spherical Bessel function. Show that

$$\int_{-1}^1 e^{ixy} P_m(y)\,dy = 2i^m j_m(x).$$

7.8-8 Let $1+\alpha, 1+\beta \in \mathbb{C}_>$ and $r, s \in \mathbb{C}$. Assume $r \neq s$ and let $y \in \mathbb{C} - [r, s]$. Let $m \in \mathbb{N}$ and define I_m by (7.8-2). Show that the Jacobi polynomial and function of the second kind are related by

$$I_m q_m(y) = \int_r^s p_m(x)(y-x)^{-1}(x-r)^\beta (s-x)^\alpha\,dx,$$

where $\mathrm{ph}(x-r) = \mathrm{ph}(s-r) = \mathrm{ph}(s-x)$. Deduce that $I_m q_m - p_m I_0 q_0$ is a polynomial of degree $m-1$.

7.8-9 Let $\nu \in \mathbb{C}_>$, $0 \le r < 1$, and $0 \le \theta \le \pi$. Let

$$f(\cos\theta) = \sum_{n=0}^\infty a_n C_n^\nu(\cos\theta),$$

and assume that the series converges uniformly on $0 \le \theta \le \pi$. Show that

$$\sum_{n=0}^\infty a_n r^n C_n^\nu(\cos\theta) = \frac{1-r^2}{\mathrm{B}(\nu+\frac{1}{2}, \frac{1}{2})}$$
$$\cdot \int_0^\pi f(\cos\varphi) R_{-\nu-1}[\nu, \nu; 1-2r\cos(\theta+\varphi)+r^2, 1-2r\cos(\theta-\varphi)+r^2](\sin\varphi)^{2\nu}\,d\varphi.$$

If $2\nu - 1 \in \mathbb{N}$, this is the solution of Laplace's equation inside the unit sphere in $\mathbb{R}^{2\nu+2}$ with boundary values f (see Section 6.7).

7.8-10 Let $n \in \mathbb{N}$, $\nu + \frac{1}{2} \in \mathbb{C}_>$, and $A, B \in \mathbb{C}$. Let f be holomorphic on an open elliptic disk with foci $A+B$ and $A-B$. Show that

$$\int_{-1}^1 f(A+Bx)C_n^\nu(x)(1-x^2)^{\nu-1/2}\,dx$$
$$= \frac{\pi^{1/2}\Gamma(\nu+\frac{1}{2})(2\nu, n)}{\Gamma(\nu+1+n)n!}\left(\frac{1}{2}B\right)^n F^{(n)}\left(\frac{1}{2}+\nu+n, \frac{1}{2}+\nu+n; A+B, A-B\right).$$

7.9-1 Let $\beta, t, x \in \mathbb{C}$. Assume $|t| < 1$. Show that

$$(1-t)^{-1-\beta}\exp\left(\frac{tx}{t-1}\right) = \sum_{n=0}^{\infty} t^n L_n^{\beta}(x).$$

7.9-2 Let $1+\beta \in U$ and $\alpha, t, x \in \mathbb{C}$. Assume $|t| < 1$. Show that

$$(1-t)^{-1-\alpha}\,{}_1F_1\left(1+\alpha; 1+\beta; \frac{tx}{t-1}\right) = \sum_{n=0}^{\infty} \frac{(1+\alpha, n)}{(1+\beta, n)} t^n L_n^{\beta}(x).$$

Exercise 7.9-1 is the special case $\alpha = \beta$.

7.9-3 Let $n \in \mathbb{N}$ and $\beta, x \in \mathbb{C}$. Let γ be a circle in $\mathbb{C}$ with center 0 described counter-clockwise. Show that

$$L_n^{\beta}(x) = L_n^{\beta}(0)\frac{1}{2\pi i}\int_{\gamma} e^t t^{-n-1} R_n(1, \beta; t-x, t)\,dt.$$

7.9-4 Let $x \in \mathbb{C}$, $z \in \mathbb{C}^k$, $b \in U_k$, and $1 + \beta = \sum_{i=1}^{k} b_i$. Assume $|z| < 1$. Show that

$$\prod_{i=1}^{k}(1-z_i)^{-b_i} S\left(b, \frac{xz}{z-1}\right) = \sum_{n=0}^{\infty} L_n^{\beta}(x) R_n(b, z),$$

where $xz/(z-1)$ is the k-tuple with ith component $xz_i/(z_i - 1)$. Exercise 7.9-1 is the case $z = (t, \ldots, t)$, and Exercise 7.9-2 is the case $z = (t, 0, \ldots, 0)$.

7.9-5 Let $1+\beta \in U$ and $t, x \in \mathbb{C}$. Show that

$$e^t\,{}_0F_1(1+\beta; -tx) = \sum_{n=0}^{\infty} \frac{t^n}{(1+\beta, n)} L_n^{\beta}(x).$$

7.9-6 Let $n \in \mathbb{N}$ and $\alpha, \beta, x \in \mathbb{C}$. Show that

$$L_n^{\alpha+\beta}(x) = \sum_{m=0}^{n} \frac{(\alpha, m)}{m!} L_{n-m}^{\beta}(x).$$

7.9-7 Let $n \in \mathbb{N}$ and $\alpha, \beta, x, y \in \mathbb{C}$. Prove the addition theorem

$$L_n^{\alpha+\beta+1}(x+y) = \sum_{m=0}^{n} L_m^{\alpha}(x) L_{n-m}^{\beta}(y).$$

Exercise 7.9-6 is the case $x = 0$.

7.9-8 Let $n - 1 \in \mathbb{N}$ and $\beta \in \mathbb{C}$. Show that

$$\frac{d}{dx} L_n^{\beta}(x) = -L_{n-1}^{\beta+1}(x).$$

7.9-9 Let $n \in \mathbb{N}$ and $n - m \in \mathbb{N}$. Let $\alpha, \beta, x \in \mathbb{C}$ and assume $(1+\alpha+\beta, n) \neq 0$. Define $f(x) = L_n^{\alpha}(x)$ and show that the Dirichlet average of $f^{(m)}$ satisfies

$$F^{(m)}(1+\alpha+m, \beta; x, 0) = (-1)^m \frac{(1+\alpha+m, n-m)}{(1+\alpha+\beta+m, n-m)} L_{n-m}^{\alpha+\beta+m}(x).$$

7.10-1 Let $n \in \mathbb{N}$. Since the frequency function of the normal distribution is even, its odd moments vanish. Show that the even moments are given by

$$\int_{\mathbb{R}} x^{2n}\,d\gamma(x) = \left(\frac{1}{2}, n\right) \qquad \text{or} \qquad (2\pi)^{-1/2}\int_{\mathbb{R}} x^{2n} e^{-x^2/2}\,dx = 2^n\left(\frac{1}{2}, n\right).$$

The variance ($n = 1$) is $\frac{1}{2}$ in one case and 1 in the other.

7.10-2 Let $n \in \mathbb{N}$ and $x \in \mathbb{C}$. Show that the monic Hermite polynomial is the result of averaging z^n with respect to the measure γ over a line through x parallel to the imaginary axis:

$$\tilde{p}_n(x) = \int_{\mathbb{R}} (x + it)^n \, d\gamma(t).$$

7.10-3 Let $n \in \mathbb{N}$ and $x \in \mathbb{C}$. Show that

$$\tilde{p}_n(x) = \int_0^\infty R_n\left(\frac{1}{2}, \frac{1}{2}; x + i\sqrt{t}, x - i\sqrt{t}\right) e^{-t} \, dt.$$

7.10-4 Let $t, x \in \mathbb{C}$. Prove the generating relation,

$$e^{2xt - t^2} = \sum_{n=0}^\infty \frac{t^n 2^n}{n!} \tilde{p}_n(x) = \sum_{n=0}^\infty \frac{t^n}{n!} H_n(x).$$

7.10-5 Let $n \in \mathbb{N}$ and $x \in \mathbb{C}$. If C_n^ν is a Gegenbauer polynomial, show that

$$H_n(x) = n! \lim_{\nu \to \infty} \nu^{-n/2} C_n^\nu(x\nu^{-1/2}).$$

7.10-6 Let $n \in \mathbb{N}$ and $x, y, \theta \in \mathbb{C}$. Prove the addition theorem,

$$H_n(x \cos\theta + y \sin\theta) = \sum_{m=0}^n \binom{n}{m} (\cos\theta)^m (\sin\theta)^{n-m} H_m(x) H_{n-m}(y).$$